



Camada Limite Planetária **Yonsei University (YSU)**



Equações governantes para o escoamento médio da atmosférico

Equações Conservação de Momentum

$$\begin{aligned} \frac{\partial(\bar{u})}{\partial t} + (\bar{u}) \frac{\partial(\bar{u})}{\partial x} + (\bar{v}) \frac{\partial(\bar{u})}{\partial y} + (\bar{w}) \frac{\partial(\bar{u})}{\partial z} + \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x} - 2\Omega\eta_3(\bar{v}) - \nu \frac{\partial^2(\bar{u})}{\partial x^2} - \nu \frac{\partial^2(\bar{u})}{\partial y^2} - \nu \frac{\partial^2(\bar{u})}{\partial z^2} &= -\frac{\partial(\overline{u'u'})}{\partial x} - \frac{\partial(\overline{v'u'})}{\partial y} - \frac{\partial(\overline{w'u'})}{\partial z} \\ \frac{\partial(\bar{v})}{\partial t} + (\bar{u}) \frac{\partial(\bar{v})}{\partial x} + (\bar{v}) \frac{\partial(\bar{v})}{\partial y} + (\bar{w}) \frac{\partial(\bar{v})}{\partial z} + \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial y} + 2\Omega\eta_3(\bar{u}) - \nu \frac{\partial^2(\bar{v})}{\partial x^2} - \nu \frac{\partial^2(\bar{v})}{\partial y^2} - \nu \frac{\partial^2(\bar{v})}{\partial z^2} &= -\frac{\partial(\overline{u'v'})}{\partial x} - \frac{\partial(\overline{v'v'})}{\partial y} - \frac{\partial(\overline{w'v'})}{\partial z} \\ \frac{\partial(\bar{w})}{\partial t} + (\bar{u}) \frac{\partial(\bar{w})}{\partial x} + (\bar{v}) \frac{\partial(\bar{w})}{\partial y} + (\bar{w}) \frac{\partial(\bar{w})}{\partial z} + \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial z} + g \frac{\bar{\rho}}{\rho_0} - \nu \frac{\partial^2(\bar{w})}{\partial x^2} - \nu \frac{\partial^2(\bar{w})}{\partial y^2} - \nu \frac{\partial^2(\bar{w})}{\partial z^2} &= -\frac{\partial(\overline{u'w'})}{\partial x} - \frac{\partial(\overline{v'w'})}{\partial y} - \frac{\partial(\overline{w'w'})}{\partial z} \end{aligned}$$

Equação Conservação de Energia

$$\frac{\partial(\bar{T})}{\partial t} + (\bar{u}) \frac{\partial(\bar{T})}{\partial x} + (\bar{v}) \frac{\partial(\bar{T})}{\partial y} + (\bar{w}) \frac{\partial(\bar{T})}{\partial z} - S_P \bar{\omega} = -\frac{\partial(\overline{u'T'})}{\partial x} - \frac{\partial(\overline{v'T'})}{\partial y} - \frac{\partial(\overline{w'T'})}{\partial z} + \frac{\bar{J}}{C_p}$$

Equação Conservação de Massa

$$\frac{\partial(\bar{c})}{\partial t} + (\bar{u}) \frac{\partial(\bar{c})}{\partial x} + (\bar{v}) \frac{\partial(\bar{c})}{\partial y} + (\bar{w}) \frac{\partial(\bar{c})}{\partial z} = -\frac{\partial(\overline{u'c'})}{\partial x} - \frac{\partial(\overline{v'c'})}{\partial y} - \frac{\partial(\overline{w'c'})}{\partial z} + \bar{S}_c$$

$$\begin{aligned} u_3 &= w \\ x_3 &= z \\ \nu \frac{\partial^2(\bar{u}_3)}{\partial x_3^2} &\cong 0 \end{aligned}$$



Equações governantes para o escoamento médio da atmosférico

$$\frac{\partial(\bar{u})}{\partial t} = - \frac{\partial(\overline{w'u'})}{\partial z}$$

$$\frac{\partial(\bar{v})}{\partial t} = - \frac{\partial(\overline{w'v'})}{\partial z}$$

$$\frac{\partial(\bar{w})}{\partial t} = - \frac{\partial(\overline{w'w'})}{\partial z} = - \frac{\partial(\overline{w'^2})}{\partial z} = - \frac{\partial(TKE)_k}{\partial z}$$

$$\frac{\partial(\bar{T})}{\partial t} = - \frac{\partial(\overline{w'T'})}{\partial z}$$

$$\frac{\partial(\bar{c})}{\partial t} = - \frac{\partial(\overline{w'c'})}{\partial z}$$

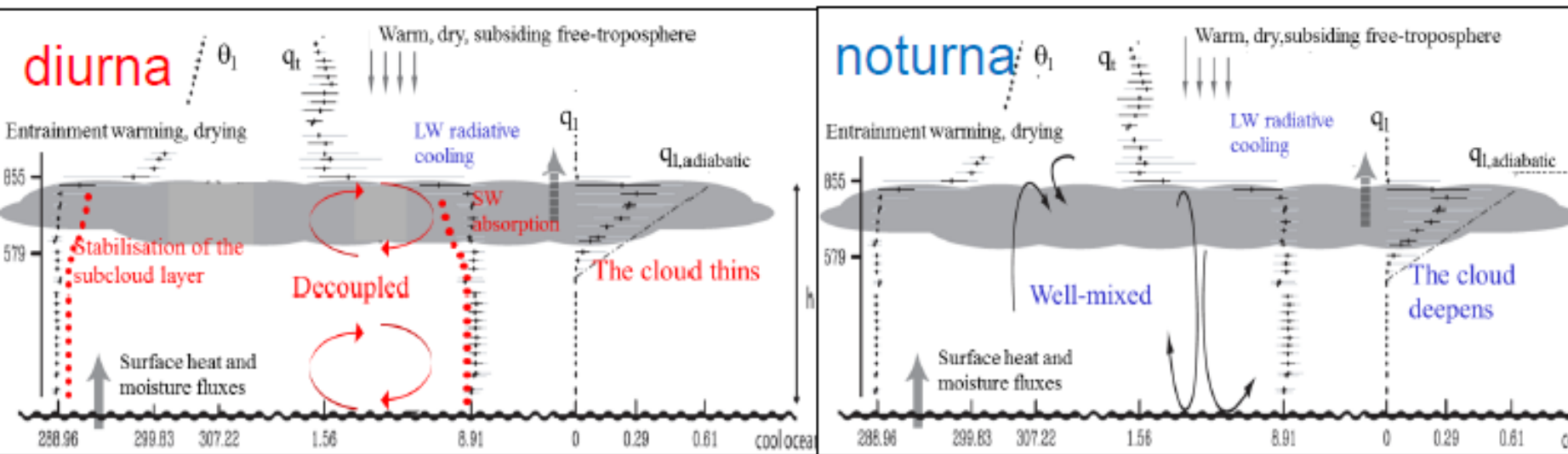
$$\frac{\partial(\overline{w'X'})}{\partial z}$$

**Mudança de fase e
processos turbulentos
presentes na atmosfera**

$$\frac{D\bar{e}}{Dt} = - \frac{(\overline{u_j'u_i'})}{2} \frac{\partial(\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i'\rho'}}{2\rho_0} \delta_{i3} - \frac{\partial(\overline{eu_j'})}{\partial x_j} - \frac{1}{2\rho_0} \frac{\partial(\overline{P'u_i'})}{\partial x_i} - \epsilon$$

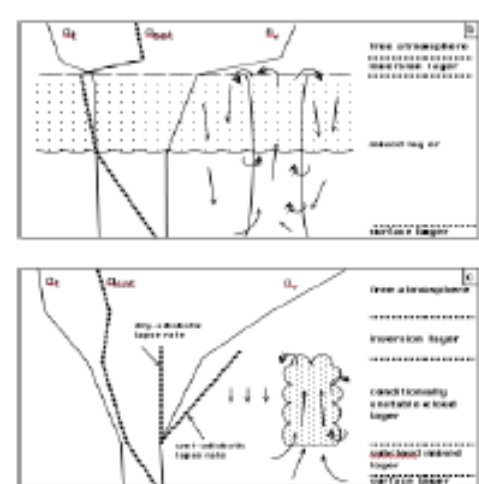
(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

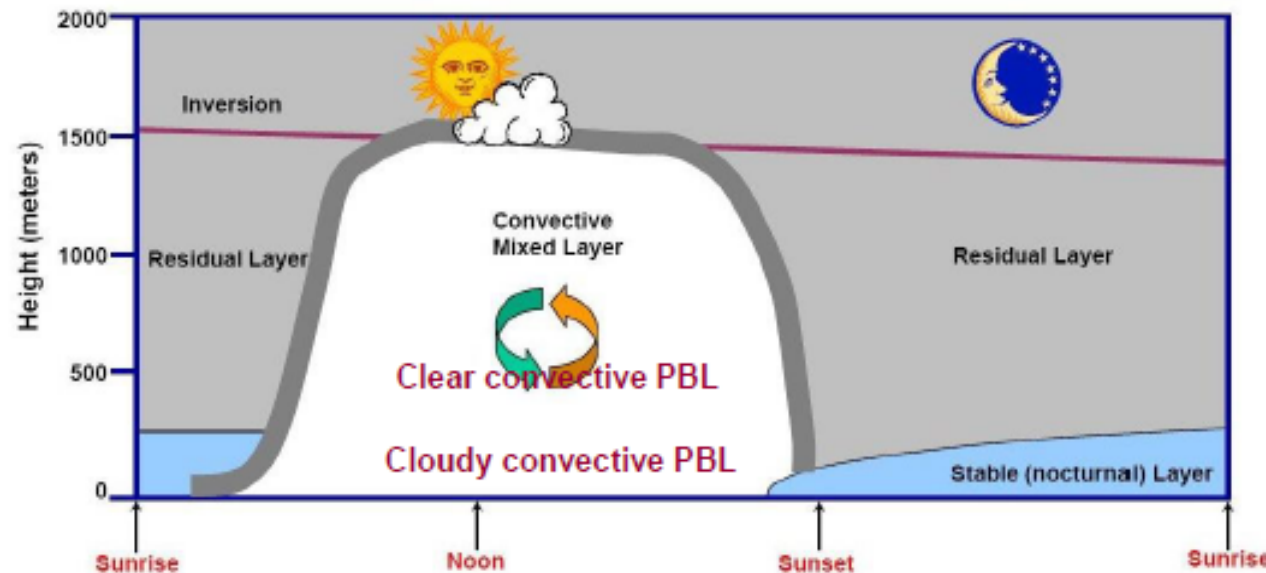


Como calcular o coeficiente de difusão turbulenta?
 $K(x, y, z, ri, cloud, rad, entr, \dots)$

Estratocumulos PBL



Cumulos PBL





Camada Limite Planetária

Yonsei University (**YSU**)

"Fechamento de primeira ordem;

Conceito de Troen e Mahrt (1986) de incorporar um termo de **correção contra o gradiente** na difusão descendente expressa exclusivamente pela mistura local.

O YSU representa **explicitamente a entranhamento** no topo da CAMADA LIMITE PLANETÁRIA.“



(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite



Vantagens

"Simula de forma **mais precisa a profundidade a mistura vertical** em PBL **impulsionadas pela flutuabilidade**, e **mistura mais rasa** em regimes de ventos fortes em comparação com o (Hong et al. 2006)."

Desvantagens

"Ainda foi constatado que **aprofunda excessivamente a PBL** em ambientes de convecção **profunda** na primavera, resultando em **ar muito seco próximo à superfície** e **subestimação da CAPE** em relação a ambientes de convecção profunda (Coniglio et al. 2013)."

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

Esquema de difusão vertical da camada limite NonLocal

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial z} \left[K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) \right]$$

1. K_c é **coeficiente de difusividade de Eddy**.

2. γ_c é uma **correção para o gradiente local** que incorpora a **contribuição de eddies de grande escala ao fluxo total**

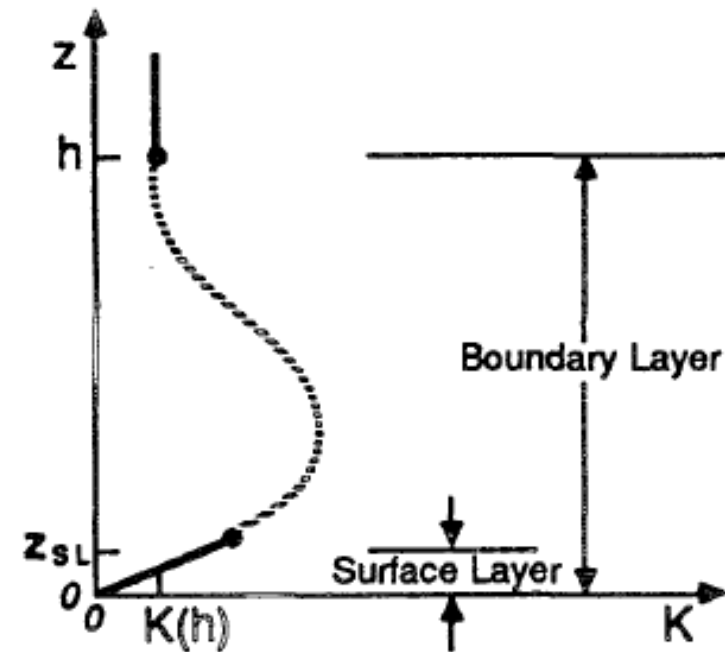


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

Difusão da camada de mistura

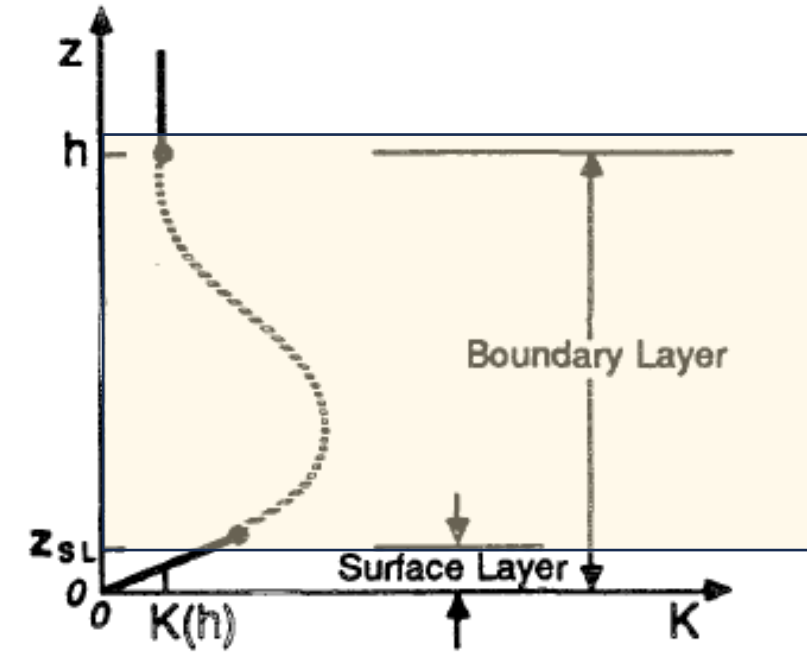


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Difusão da camada de mistura

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

$$K_{zm} = kw_s z \left(1 - \frac{z}{h}\right)^p$$

"onde:

1. p é o expoente da forma do perfil, considerado como 2.
2. k é a constante de von Kármán (= 0.4),
3. z é a altura a partir da superfície
4. h é a altura da PBL.
5. w_s é A escala de velocidade da camada mistura é representada como"

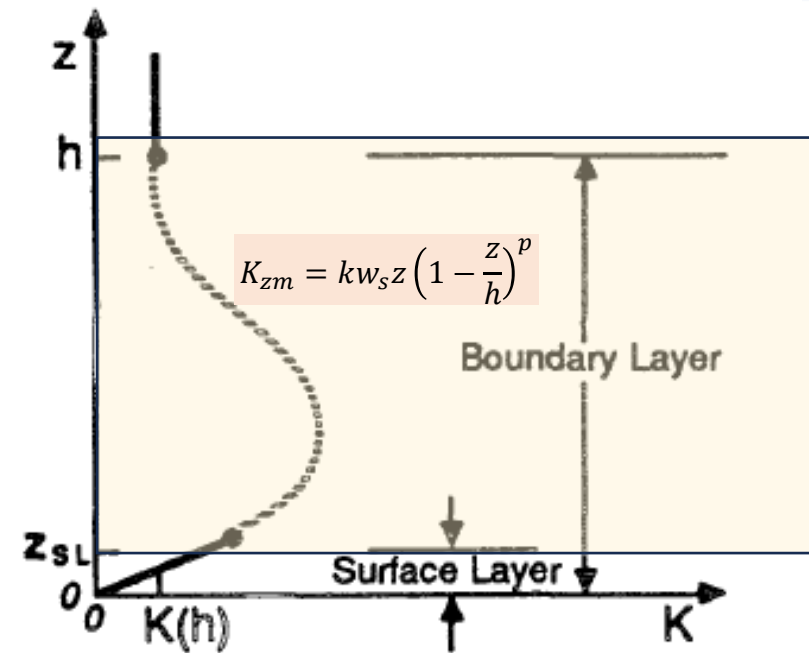


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Difusão da camada de mistura

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

Escala de velocidade da camada mistura

$$w_s = \left(u_*^3 + \frac{8kw_{*b}^3 z}{h} \right)^{\frac{1}{3}}$$

Onde:

u_* é a escala de velocidade de fricção na superfície

w_{*b} é a escala da velocidade convectiva

Escala da velocidade convectiva

$$w_{*b} = \left\{ \frac{g}{\theta_{va}} (w' \theta'_v)_0 h \right\}^{\frac{1}{3}}$$

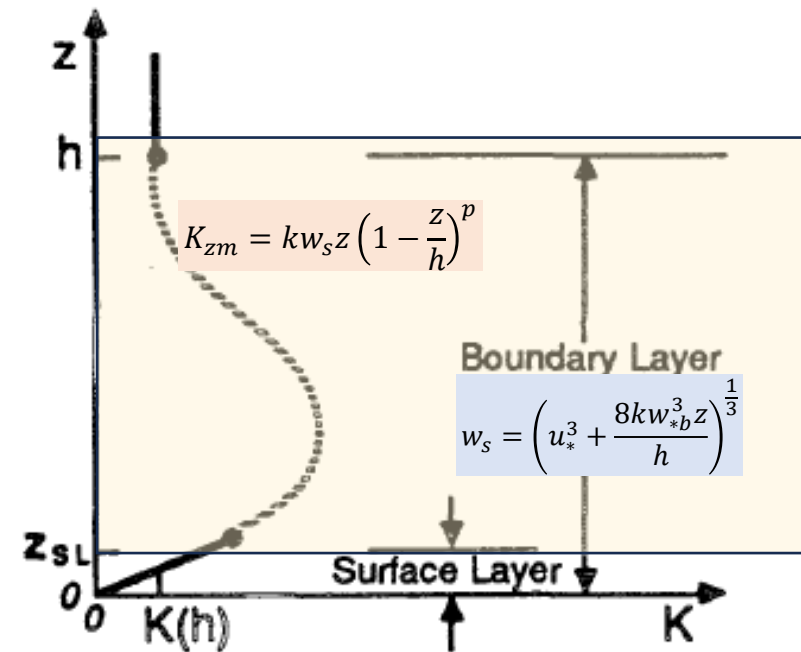


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Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

O termo de contra-gradiente para θ e momentum':

$$\gamma_c = b \frac{\overline{(w'c')}_0}{w_{s0}}$$

Onde:

$\overline{(w'c')}_0$ é fluxo de superfície correspondente a θ e a u e v .

b é um coeficiente de proporcionalidade

w_{s0} é a escala de velocidade na camada de mistura, é definida como a velocidade em $z = 0.5h$ na equação:

$$w_{s0} = w_s = \left(u_*^3 + \frac{8kw_*^3 z}{h} \right)^{\frac{1}{3}} = \left(u_*^3 + \frac{8kw_*^3 0.5h}{h} \right)^{\frac{1}{3}} = \left(u_*^3 + 4kw_*^3 \right)^{\frac{1}{3}}$$

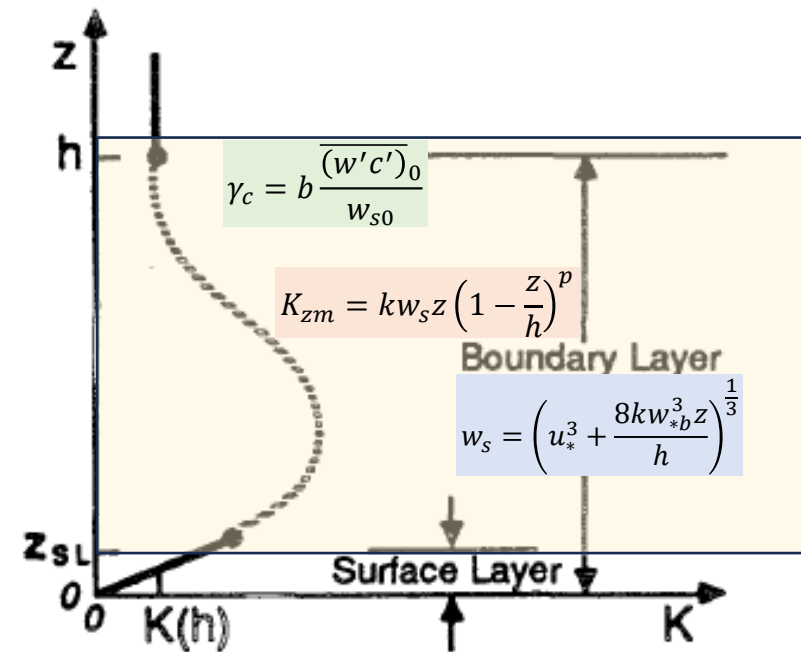


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Difusão da camada de mistura

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

"A difusividade turbulenta para temperatura e umidade K_h é calculada a partir de K_{zm} na equação (2) usando a relação do número de Prandtl."

$$P_r = \left(\frac{\phi_t}{\phi_m} + bk \frac{0.1h}{h} \right)$$

"onde:

P_r é uma constante dentro de toda a camada limite de mistura."

$$K_h \sim P_r K_{zm}$$

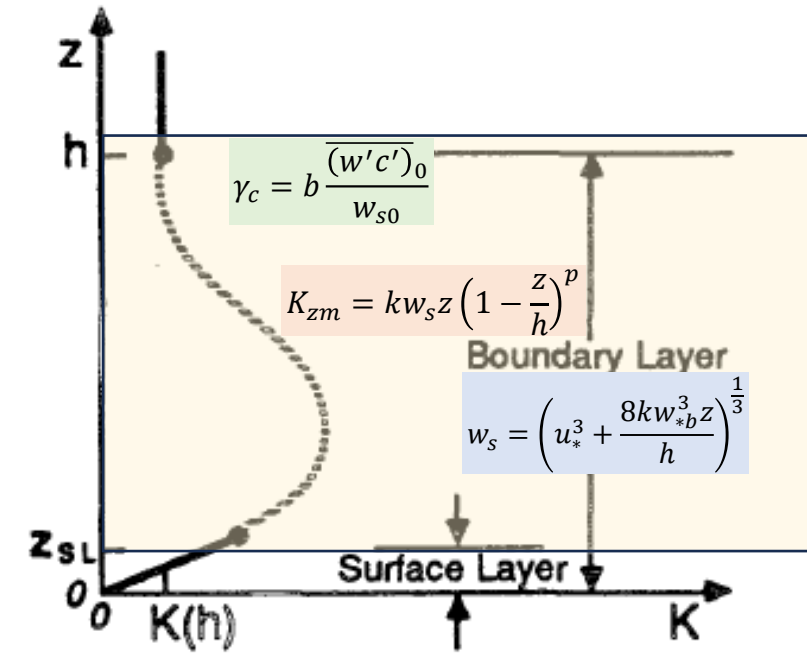


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

Difusão da camada de mistura

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

Acoplamento da Camada superficial e a camada de mistura

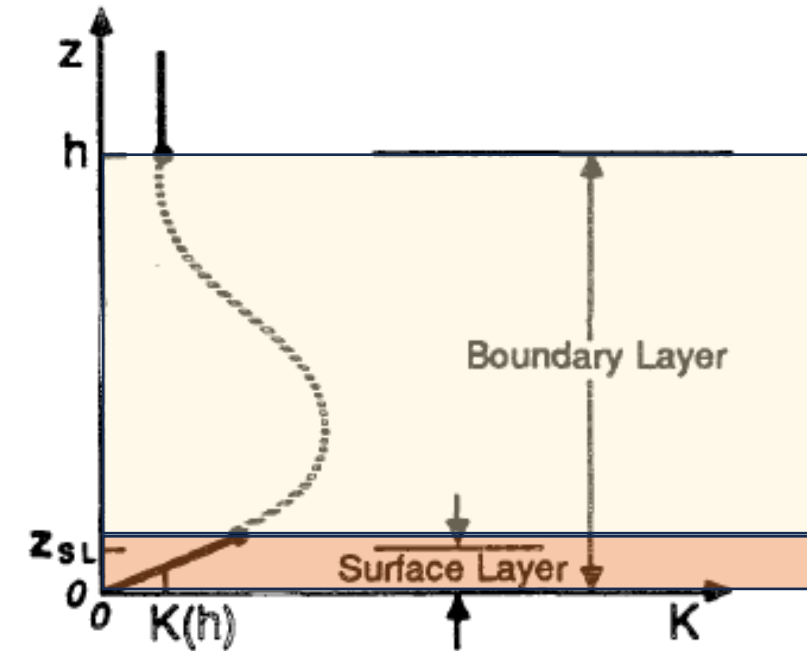


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

Camada superficial $\overline{(w'C')}_0 \sim cte$ entre $0 \leq z \leq Z_{SL}$



Como parameterizar os momentum de 2 ordem $w'\varphi'$ na **CL Superficial**



$$F = \rho K(z) \frac{d\varphi}{dz} \quad (F = \overline{w'\varphi'})$$

Formulação em diferenças finitas

$$F_{1.5} = \rho K(z_{1.5}) \frac{\varphi_2 - \varphi_1}{z_2 - z_1}$$

Integral da camada superfície :

$$\varphi_1 - \varphi_s = \int_{z_{o\varphi}}^{z_1} \frac{F_{0\varphi}}{\rho K(z)} dz$$

Camada
de fluxo
constante:

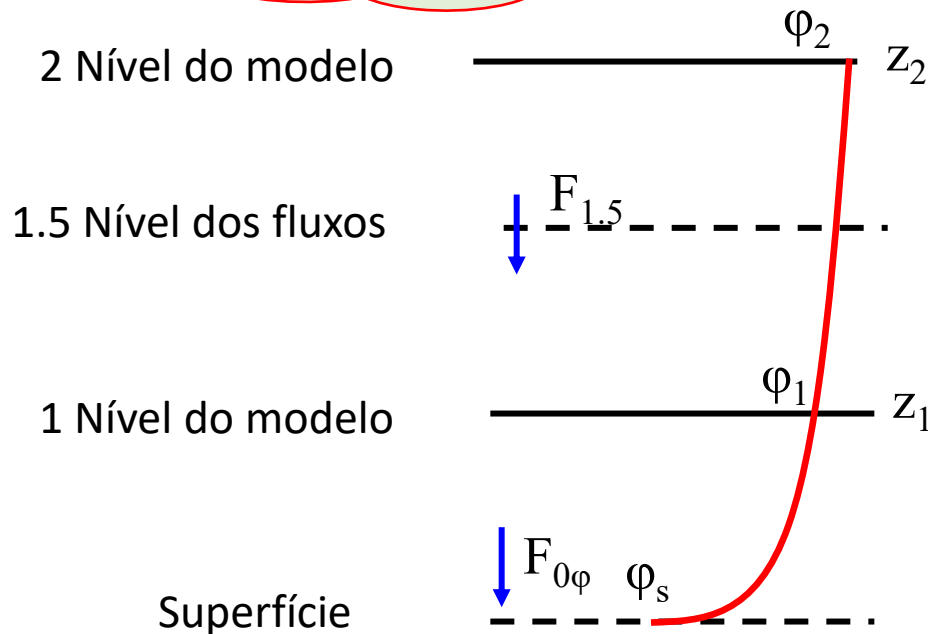
$$\varphi_1 - \varphi_s \approx \frac{F_0}{\rho} \int_{z_{o\varphi}}^{z_1} \frac{1}{K(z)} dz$$

Escoamento
neutro: $K(z) = \kappa u_* z$

$$\varphi_1 - \varphi_s \approx \frac{F_{0\varphi}}{\rho \kappa u_*} \int_{z_{o\varphi}}^{z_1} \frac{dz}{z} \Rightarrow \varphi_1 - \varphi_s = \frac{F_{0\varphi}}{\rho \kappa u_*} \ln \left(\frac{z_1}{z_{o\varphi}} \right)$$

κ : Von Karman constant (0.4)
 u_* : Friction velocity
 ρ : Density

Calcular $K(z_{1.5})$???



u, v, T, q

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

Função universal integrada para a camada limite superficial

$$(z_0 \leq z \leq z_{SL})$$

when $\zeta < 0$ (unstable)

$$\Psi_M = \ln \left(\frac{z_m - d_0}{z_0} \right) + \ln \frac{(x_0^2 + 1)(x_0 + 1)^2}{(x^2 + 1)(x + 1)^2} + 2(\tan^{-1} x - \tan^{-1} x_0) \quad (2.59)$$

$$\Psi_H = \ln \left(\frac{z_m - d_0}{z_0} \right) + 2 \ln \left(\frac{y_0 + 1}{y + 1} \right) \quad (2.60)$$

$$x = (1 - 16\zeta)^{1/4}, \quad x_0 = (1 - 16\zeta_0)^{1/4}, \quad y = (1 - 16\zeta)^{1/2}, \quad y_0 = (1 - 16\zeta_0)^{1/2} \quad (2.61)$$

when $\zeta \geq 0$ (stable)

$$\Psi_M = \ln \left(\frac{z_m - d_0}{z_0} \right) + \frac{7}{3} \ln \frac{1 + 3\zeta + 10\zeta^3}{1 + 3\zeta_0 + 10\zeta_0^3} \quad (2.62)$$

$$\Psi_H = \ln \left(\frac{z_m - d_0}{z_0} \right) + 400 \ln \frac{1 + 7/400\zeta + 0.005\zeta^2}{1 + 7/400\zeta_0 + 0.005\zeta_0^2} \quad (2.63)$$

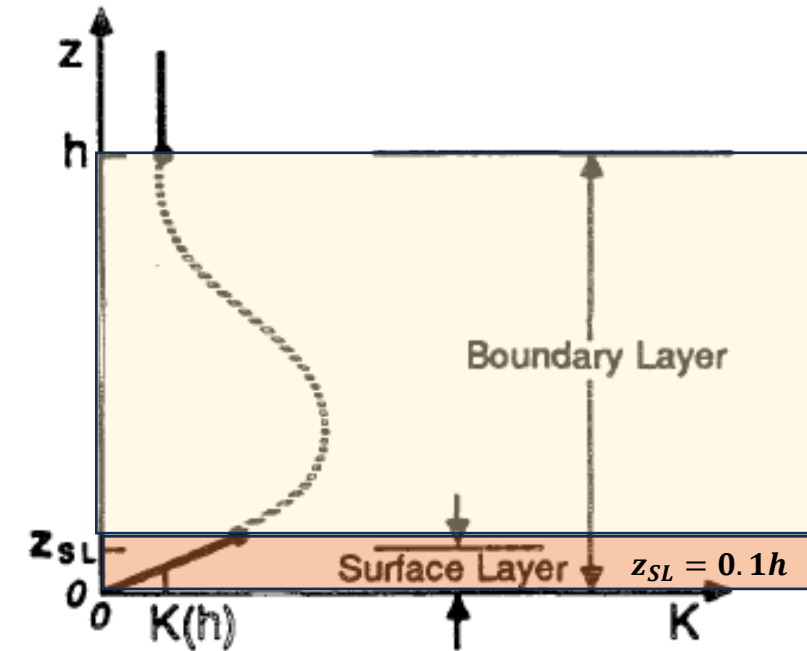


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

$$H = \rho_a C_p (T_{au} - T_m) \kappa u_* / \Psi_H \quad (2.56)$$

$$L = -\rho_a C_p T_m u_*^3 / \kappa g H \quad (2.57)$$

$$\zeta = z_m / L \quad (2.58)$$

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

“Para satisfazer a compatibilidade entre o topo da camada superficial ($z = 0.1h$) e a base da PBL .

$$C_{DN} = \frac{k^2}{\left[\ln \left(\frac{z_r}{z_0} \right) \right]^2} \quad \text{condições neutras}$$

$$F = \rho K(z) \frac{d\varphi}{dz} \quad (F = \overline{w'\varphi'})$$

$$\therefore \frac{1}{r_{aw}} = C_D U_r$$

$$\begin{aligned} F_{0\varphi} &= \tau = H = LE \\ \tau &= \frac{1}{r_{aw}} \rho \mathbf{u}_r \\ H &= \frac{1}{r_{aw}} \rho c_p (T - T_r) \\ LE &= \frac{1}{(r_{aw} + r_{soil})} (\rho c_p / \gamma) [h e_s(T) - e_r] \end{aligned}$$

$$\begin{aligned} \tau_{wb} &= \rho u_*^2 \\ r_{aw} &= \frac{\Psi_H}{\kappa u_*} = \frac{\Psi_M \Psi_H}{\kappa^2 u_m} \end{aligned}$$

$$\begin{aligned} F_{0\varphi} &= \tau = H = LE \\ \tau &= C_D \rho \mathbf{u}_r \\ H &= C_D \rho c_p (T - T_r) \\ LE &= C_{Dh} (\rho c_p / \gamma) [h e_s(T) - e_r] \end{aligned}$$

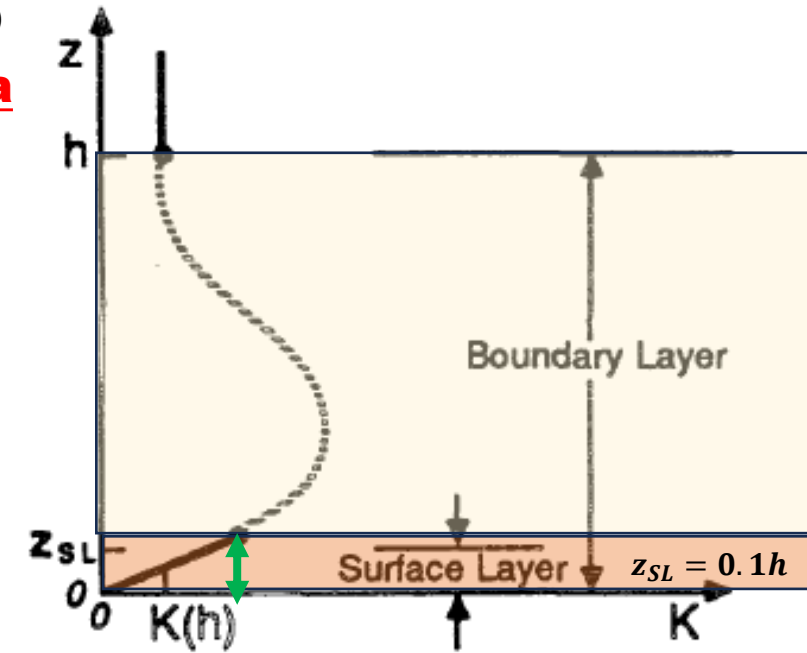


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

$$\varphi_1 - \varphi_s = \frac{F_{0\varphi}}{\rho \kappa u_*} \ln \left(\frac{z_1}{z_{0\varphi}} \right)$$

$\uparrow$
 u, v, T, q

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

“Para satisfazer a compatibilidade entre o topo da camada superficial ($z = 0.1h$) e a base da PBL .

Função universal integrada para a camada limite de mistura
($z_{SL} \leq z \leq h$)

Primeiro para condições instáveis e neutras $\overline{(w'\theta'_v)_0} \leq 0$

$$\phi_m = \left(1 - 16 \frac{z}{L}\right)^{-\frac{1}{4}} = \left(1 - 16 \frac{0.1h}{L}\right)^{-\frac{1}{4}}, \text{ para } u \text{ e } v$$

$$\phi_t = \left(1 - 16 \frac{z}{L}\right)^{-\frac{1}{2}} = \left(1 - 16 \frac{0.1h}{L}\right)^{-\frac{1}{2}}, \text{ para } \theta \text{ e } q$$

Emquanto para regime estável $\overline{(w'\theta'_v)_0} > 0$

$$\phi_m = \phi_t = \left(1 - 5 \frac{z}{L}\right) = \left(1 - 5 \frac{0.1h}{L}\right)$$

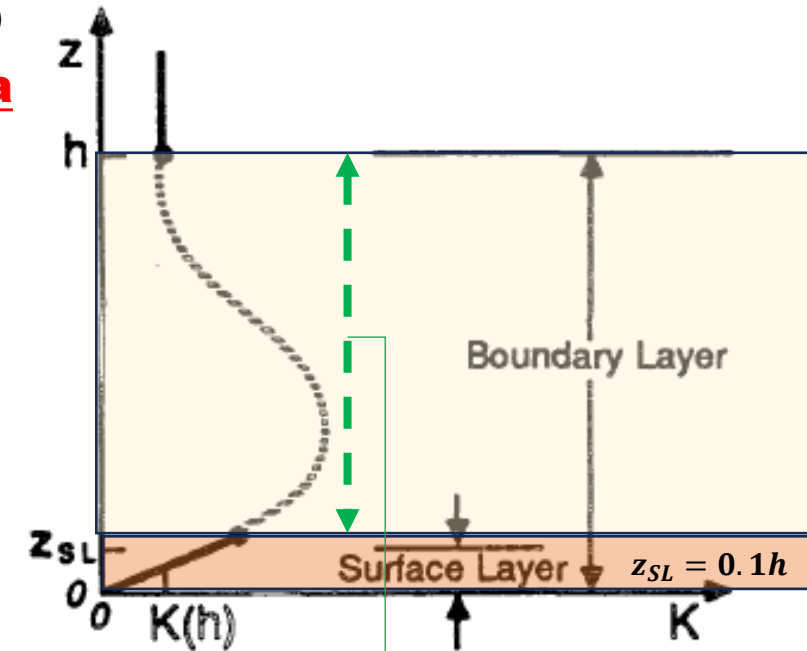


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As funções universais com perfil similares (regime de estabilidade) às da física da camada superficial devem ser usadas

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

O topo da camada superficial é estimada em $0.1h$ para estimar o fator b da equação do contra-gradiente, o expoente $-\frac{1}{3}$ é escolhido para assegurar o limite de convecção livre. Portanto, utiliza-se a aproximação:

$$\phi_m = \left(1 - 16 \frac{0.1h}{L}\right)^{-\frac{1}{4}} \approx \left(1 - 12 \frac{0.1h}{L}\right)^{-\frac{1}{3}}, \text{ para } u \text{ e } v$$

$$w_{*b} = \left\{ \frac{g}{\theta_{va}} (w' \theta'_v)_0 h \right\}^{\frac{1}{3}}$$

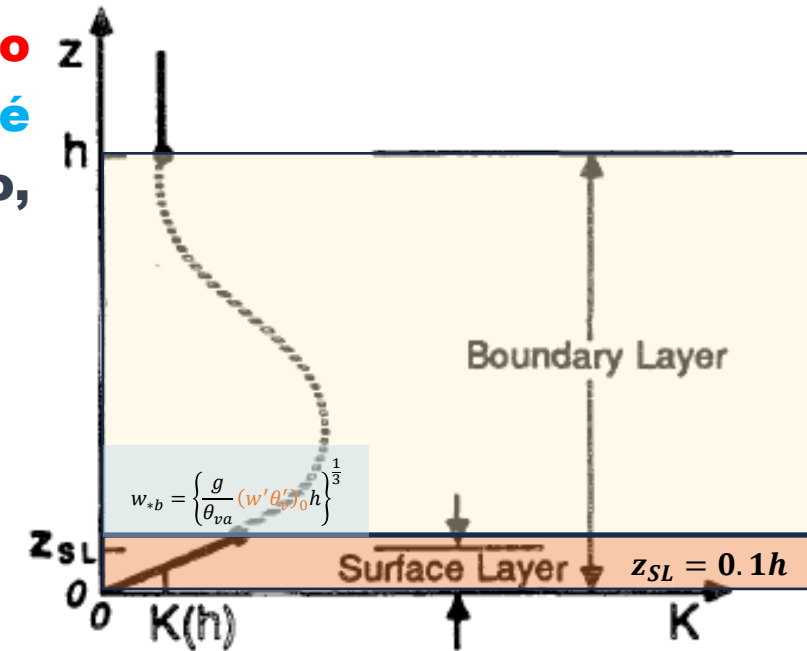


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

Baseado na derivação de Troen and Mahrt (1986) Holtslag (1990), encontra-se

$$b = 7.8$$

$$P_r = \left(\frac{\phi_t}{\phi_m} + bk \frac{0.1h}{h} \right)$$

$$K_h \sim P_r K_{zm}$$



(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

A altura da Camada Limite Planetária é definida:

$$h = Rib_{cr} \frac{\theta_{va} |U(h)|^2}{g(\theta_v(h) - \theta_s)}$$

Onde:

Rib_{cr} é o numero de Richardson bulk crítico

$U(h)$ é a velocidade do vento horizontal em h

θ_{va} é a temperatura virtual no nível mais baixo do modelo (30-50 m) da superfície

$\theta_v(h)$ é a temperatura potencial virtual em h

θ_s é a temperatura potencial apropriada próxima a superfície

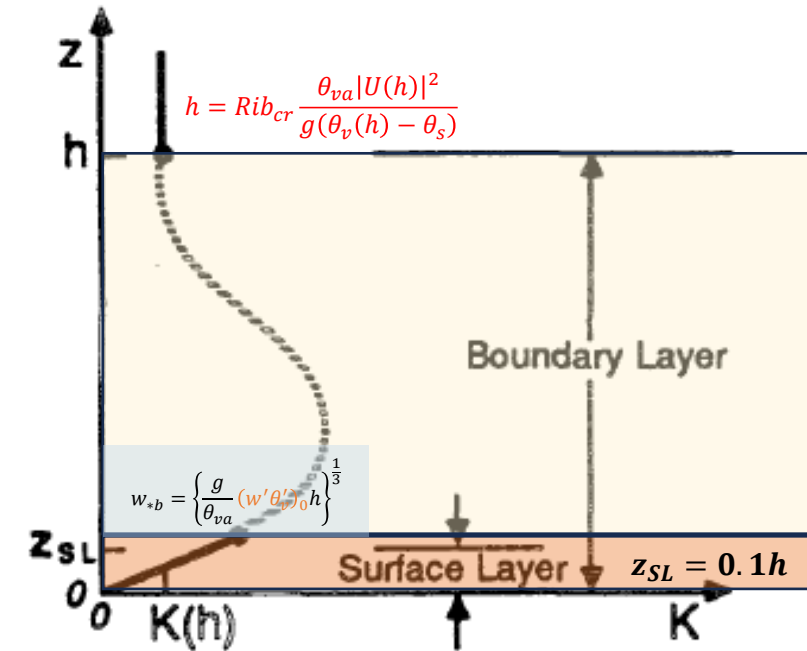


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

h é a altura da camada limite

L é a escala de comprimento de MONIN – OBUKHOV

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

Deardorff (1973), Troen and Mahrt (1986) Holtslag and Moeng (1991), Holtslag and Boville (1993)

A temperatura potencial apropriada próxima a superfície é definida:

$$\theta_s = \theta_{va} + \theta_T \left[= b \frac{\overline{(w'\theta_v')}_0}{w_s h} \right]$$

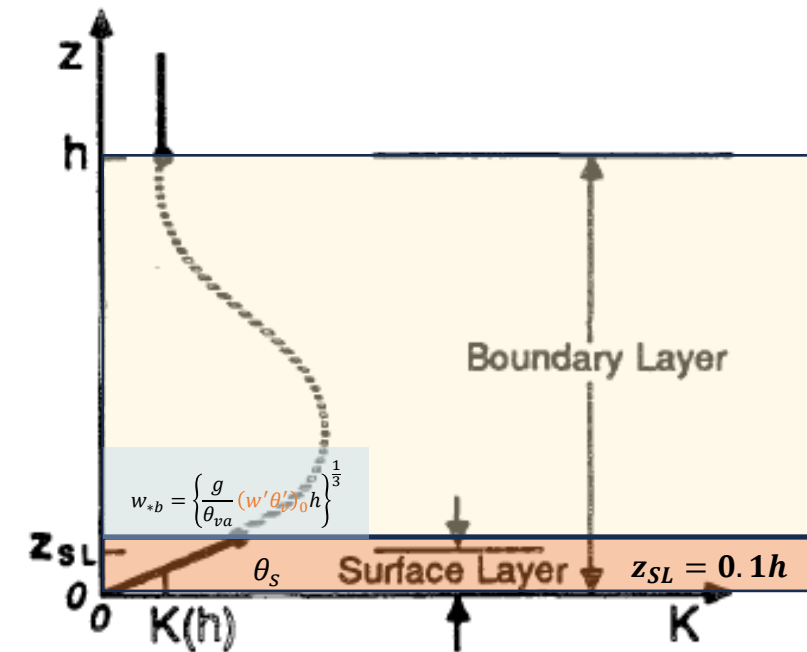


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Onde:
 θ_T é o a escala de excesso de temperatura virtual próxima a superfície

Testes preliminares indica que θ_T alguma vezes **torna muito grande quando vento de superfície é muito baixo, resultando em h irrealista**. "Este alto valor de h devido ao irrealismo de θ_T não prejudica o resultado porque resulta em um coeficiente de difusividade muito pequeno nesta situações. "mas isso não é desejável para fins de diagnóstico". Por estas razões **defini-se um limite máximo de $\theta_T = 3K$** "

Louis (1979)

Difusão na atmosfera livre

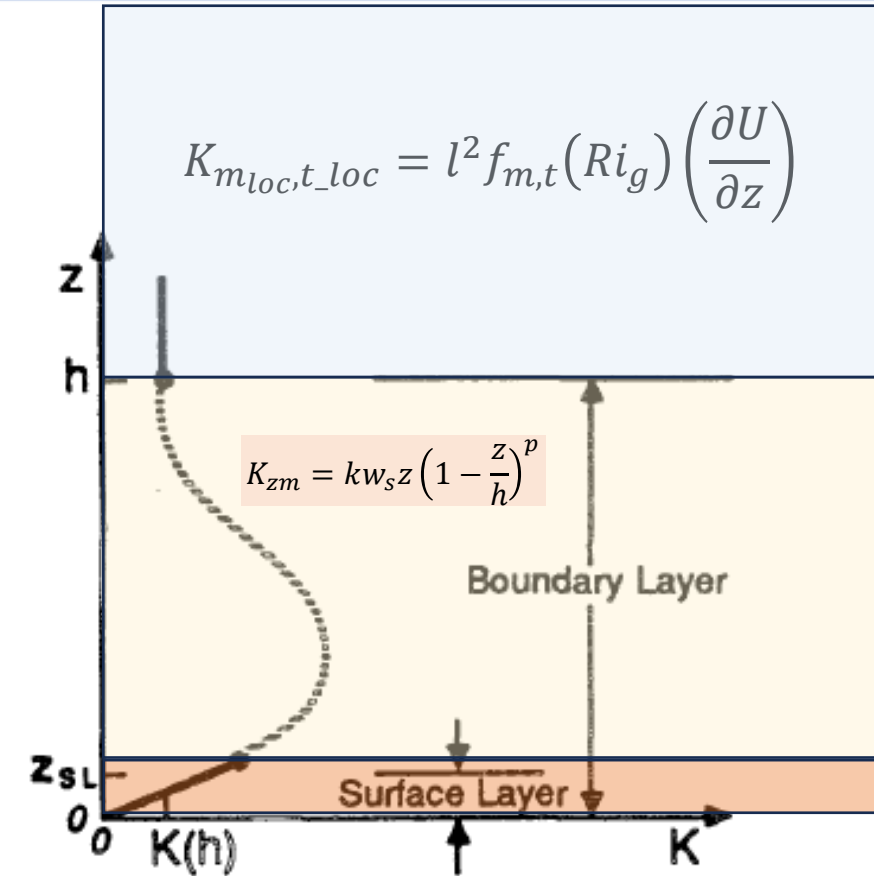


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Processos físicos da turbulência na camada limite

“Os **coeficientes de difusividade vertical para momentum** (m ; u , v) e escalares (t ; θ , q), seguindo Louis (1979) acima de h ,”

Louis (1979)

$$K_{m_{loc},t_{loc}} = l^2 f_{m,t}(Ri_g) \left(\frac{\partial U}{\partial z} \right)$$

Onde:

l é o comprimento de mistura.

$f_{m,t}(Ri_g)$ função de **estabilidade dependente do numero de Richardson bulk**.

$\frac{\partial U}{\partial z}$ **cisalhamento vertical do vento**

h é a altura da camada limite

L é a escala de comprimento de MONIN – OBUKHOV

Paulo Yoshio Kubota

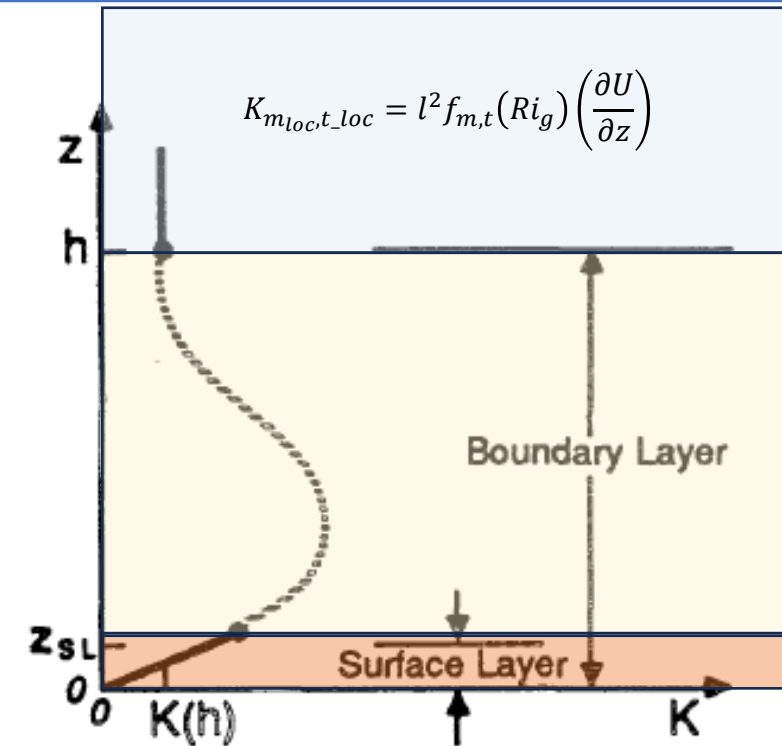


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite



Louis (1979)

“O comprimento de mistura l é dado por:,”

$$\frac{1}{l} = \frac{1}{kz} + \frac{1}{\lambda_0}$$

$$\frac{1}{l} = \frac{\lambda_0 + kz}{\lambda_0 kz}$$

Onde:

z é a altura acima da superfície.

λ_0 é a escala de comprimento assintótico ($\lambda_0=30\text{m}$)

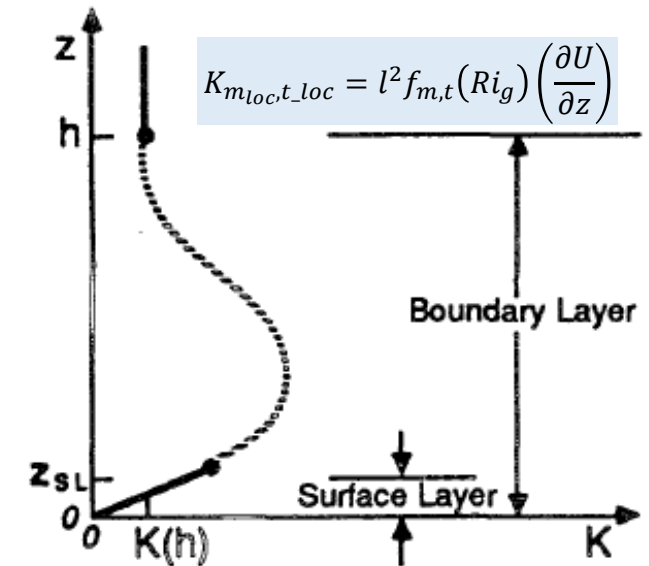
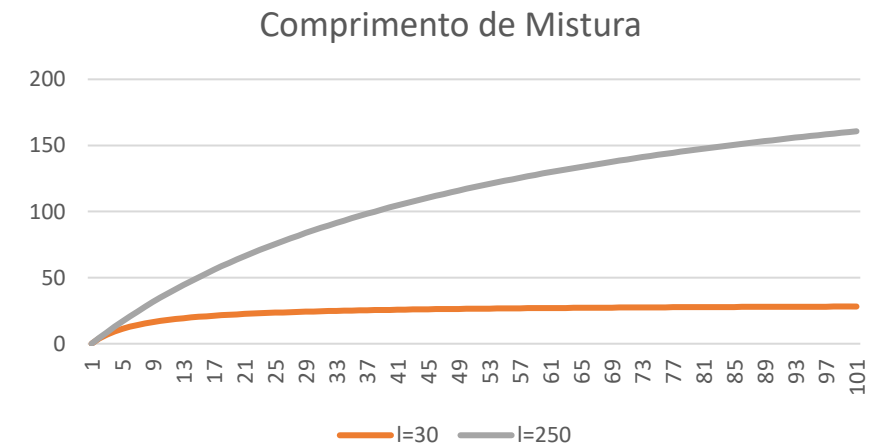


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

Kim (1991) propôs o **comprimento de mistura de 30m** para a atmosfera livre neutra através de observações por avião

No modelo operacional utiliza-se ($\lambda_0=250\text{ m}$)

$$l = \frac{\lambda_0 kz}{\lambda_0 + kz}$$



" $\frac{1}{\lambda_0} = \frac{1}{250}$ Isto é reduzido porque o regime instável dentro da camada de mistura agora é levado em consideração pelo esquema de difusão não local."

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite



A função de estabilidade $f_{m,t}(Ri_g)$ são diferentes para regimes estável e instável.

Para um atmosfera livre estavelmente estratificada $Ri_g > 0$ adota-se a equação de kim(1991)

$$f_{m,t}(Ri_g) = e^{-8.5Ri_g} + \frac{0.15}{Ri_g + 3.0}$$

Para um atmosfera livre neutralmente e instavelmente estratificada $Ri_g \leq 0$ adota-se a equação para a superfície, substituindo $\frac{0.1h}{L}$ por Ri_g

Primeiro para condições instáveis e neutras $\overline{(w'\theta_v')}_0 \leq 0$

$$\phi_m = \left(1 - 16 \frac{0.1h}{L}\right)^{-\frac{1}{4}}, \text{ para } u \text{ e } v$$

$$f_m(Ri_g) = (1 - 16Ri_g)^{-\frac{1}{4}}$$

$$\phi_t = \left(1 - 16 \frac{0.1h}{L}\right)^{-\frac{1}{2}}, \text{ para } \theta \text{ e } q$$

$$f_t(Ri_g) = (1 - 16Ri_g)^{-\frac{1}{2}}$$

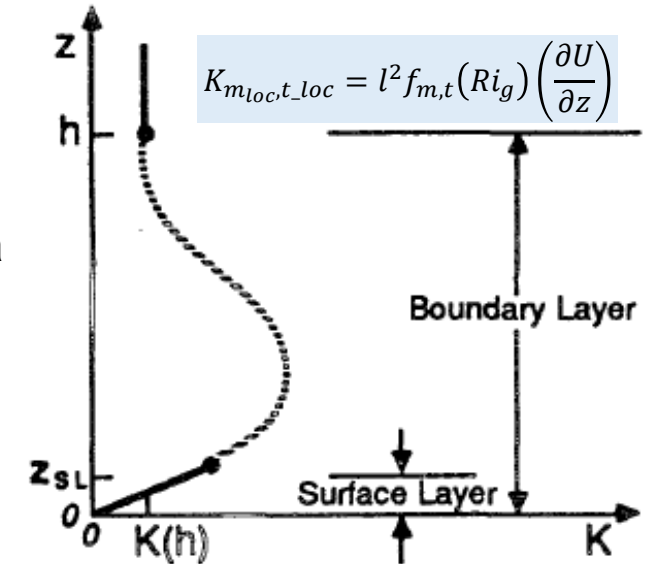


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

(1)Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

"O esquema YSU também **considera o fluxo de entranhamento acima de h** , que expressa a **penetração do fluxo de entranhamento acima de h** , **independentemente da estabilidade local**.

O algoritmo YSU-06 não contém uma formulação específica para a Camada Limite Superficial (SBL).

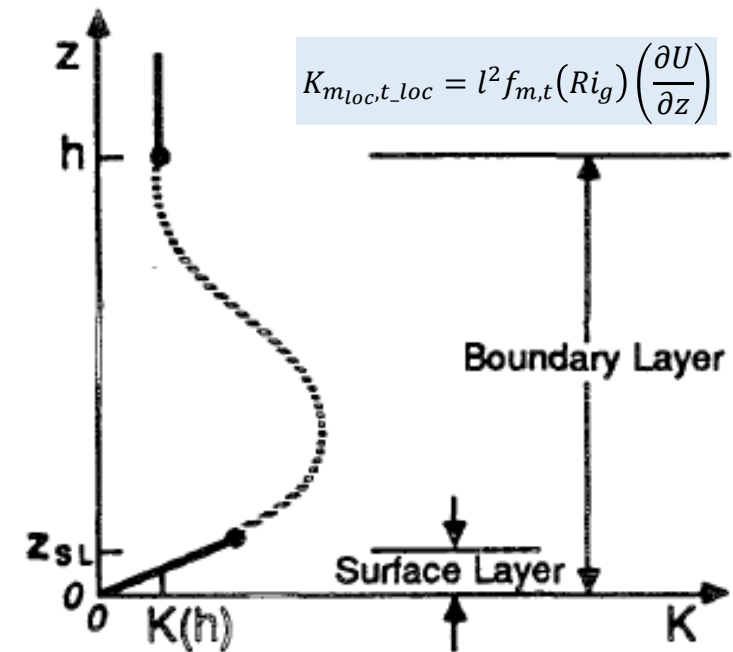


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

Em outras palavras, **a mistura de turbulência dentro da SBL é tratada como uma difusão para uma atmosférica livre**, calculando os coeficientes de difusão com:

$$\frac{1}{l} = \frac{1}{kz} + \frac{1}{\lambda_0}$$

$$K_{m_{loc},t_{loc}} = l^2 f_{m,t}(Ri_g) \left(\frac{\partial U}{\partial z} \right)$$

h é a altura da camada limite

L é a escala de comprimento de MONIN **Paulo O'Brien Kubo**

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite



$$K_{m_{loc},t_{loc}} = l^2 f_{m,t}(Ri_g) \left(\frac{\partial U}{\partial z} \right)$$

$$K_h \sim P_r K_{zm}$$

Introduz-se uma **difusão de fundo (background)** $K_{z,0} = 1 \frac{m^2}{s}$

"para **compensar a difusão numérica**, devido ao modo como $K_{m_{loc},t_{loc}}$ é calculado" Limita-se o valor entre $1 \frac{m^2}{s}$ a $1000 \frac{m^2}{s}$

E o número de Prant deve ser definido entre (0.24 a 4):

$$P_r = 1.5 + 3.08 Ri_g$$

h é a altura da camada limite

L é a escala de comprimento de MONIN-OSHKUBOVA

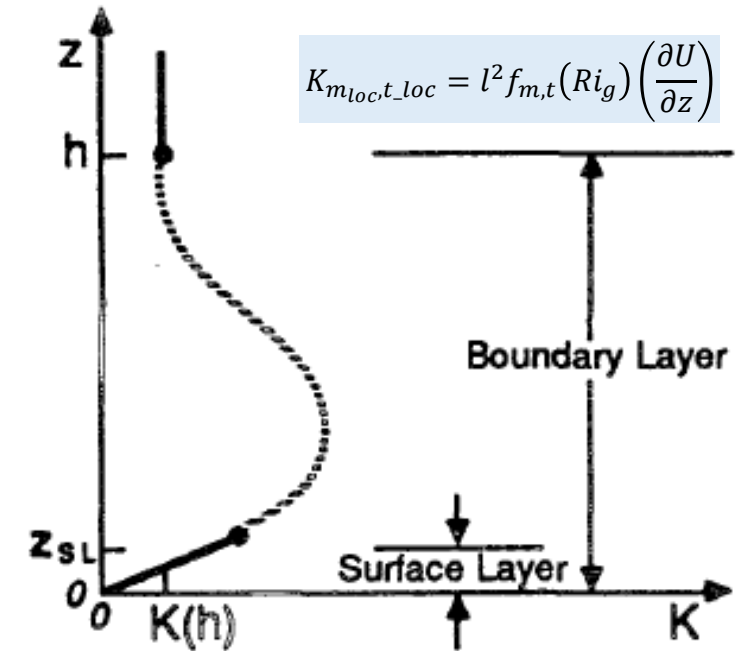


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

Implementation of a revised SBL scheme in the YSU BL package

Noh et al 2003,

“Propõem-se a adição da **contribuição do fluxo de calor devido o entranhamento no topo da Camada Limite Planetária (PBL)**”

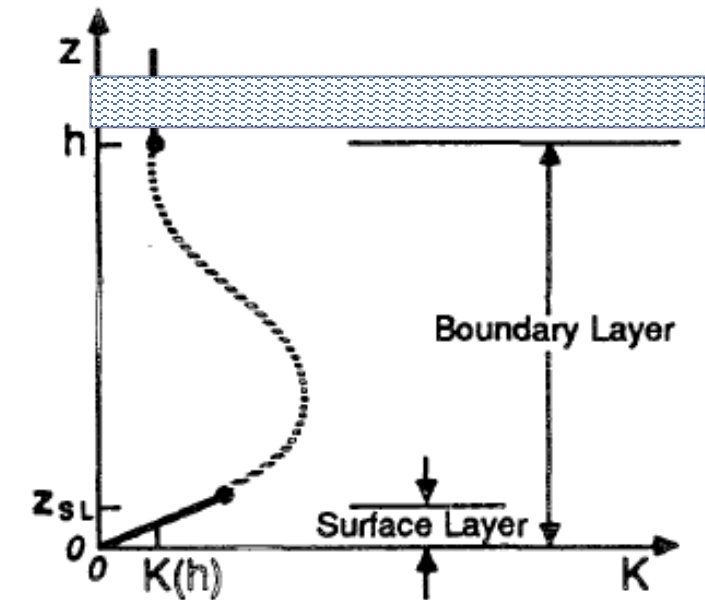


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

(1)Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

“Propõem-se a **adição da contribuição do fluxo de calor devido o entranhamento no topo da Camada Limite Planetária (PBL)**”

Noh et al 2003, modificaram o esquema original

$$-\overline{w'c'} = K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) - \overline{(w'c')}_h \left(\frac{z}{h} \right)^n$$

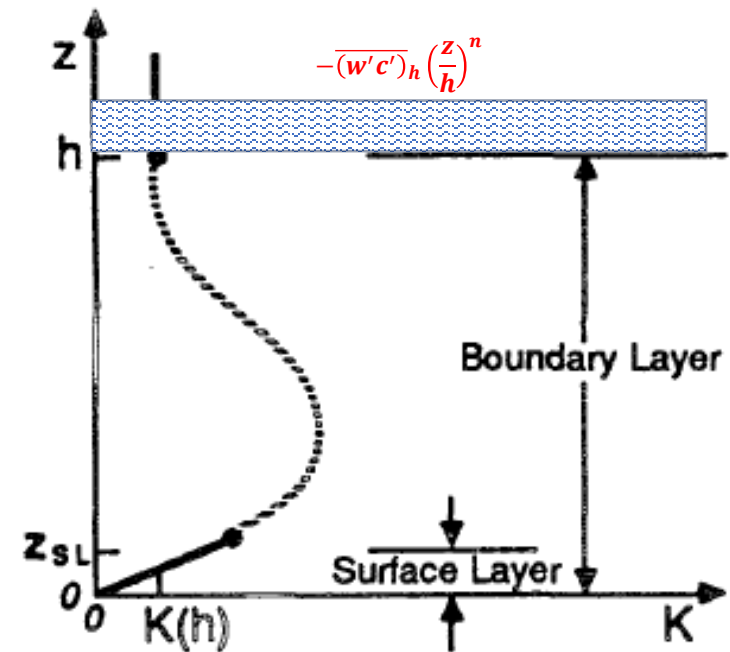


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

"Aqui obtivemos a constante empírica n como $n = 3$ com base na comparação entre os resultados do modelo e dados de LES, mas descobriu-se que os resultados são altamente insensíveis à escolha de n . Por exemplo, simulações com valores substancialmente diferentes de n , como $n = 1$ ou 5 , também apresentam resultados semelhantes, pois a contribuição do novo termo é desprezado $\left(\frac{z}{h} \right)^n$, exceto próximo ao topo da Camada Limite Planetária (PBL $z = h \Rightarrow \frac{z}{h} = 1$)."

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

"O **esquema YSU antigo** usava uma teoria K modificada, com um termo de contra-gradiente adicional que incorpora a contribuição de vórtices de grande escala ao fluxo total."

Hong SY et al 2006. **Noh et al 2003**, modificaram o esquema original introduzindo o **efeito de entranhamento**

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial z} \left\{ K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) - \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right\}$$

$\overline{(w'c')}_h$ é o fluxo na camada de inversão

"A fórmula mantém o conceito básico de Hong SY et al 1996, mas inclui um **termo assintótico de fluxo de entrada na camada de inversão** $\overline{(w'c')}_h \left(\frac{z}{h} \right)^3$."

Neste caso a altura da (PBL) h é definida como o nível em que o fluxo mínimo ocorre na camada de inversão, enquanto em Hong SY et al 1996 é definida como o nível em que a mistura turbulenta da camada limite diminui.

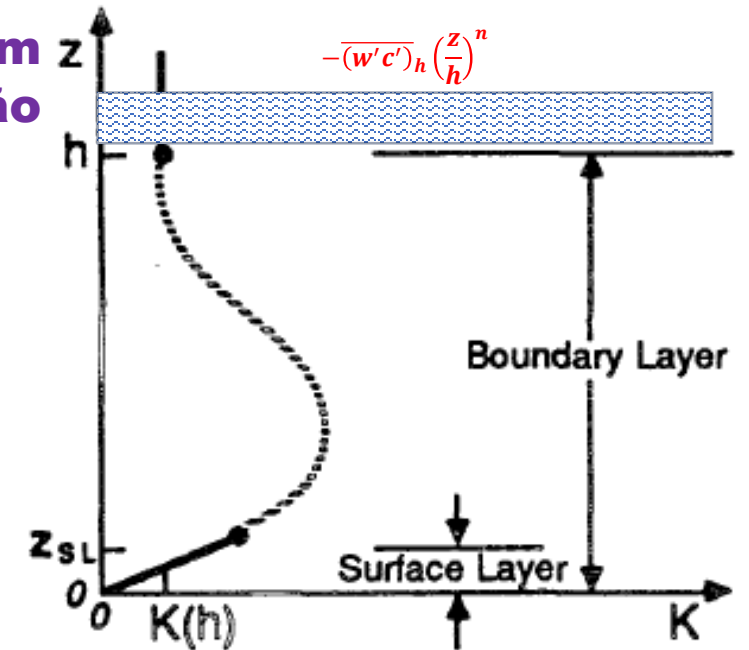


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).



(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

$\overline{(w'c')}_h$ é o fluxo na camada de inversão “em condições de convecção livre, geralmente é estimado por”

$$\overline{(w'c')}_h$$

$$\frac{\overline{w'\theta'_h}}{\overline{w'\theta'_0}} = -A_R$$

$$\overline{w'\theta'_h} = -A_R \overline{w'\theta'_0}$$

$$\overline{w'\theta'_h} = -A \frac{w_*^3}{h}$$

“e o valor apropriado para A_R ($= \left(\frac{g}{T_0}\right) A$) é sugerido estar na faixa entre 0,1 e 0,3 (Ball, 1960).”

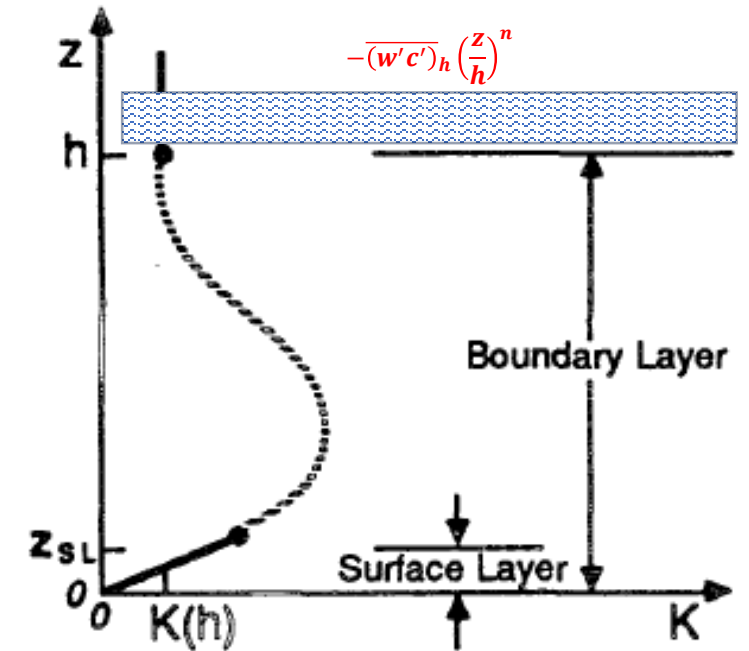


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

“Enquanto isso, na presença de cisalhamento, Moeng e Sullivan (1994) e Driedonks (1982) estenderam a fórmula ”

$$\overline{w'\theta'_h} = -A \frac{w_m^3}{h}$$

$$w_m^3 = w_*^3 + B u_*^3$$

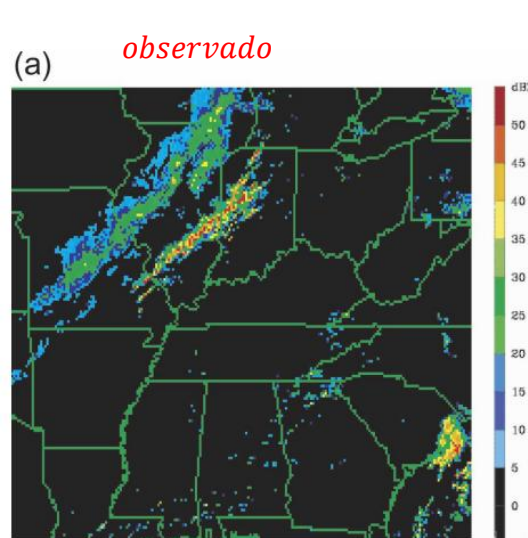
“com o valor sugerido $B = 5$ (Moeng e Sullivan, 1994) ou $B = 25$ (Driedonks, 1982).

$$w_s = \left(u_*^3 + \frac{8kw_*^3 z}{h} \right)^{\frac{1}{3}}$$

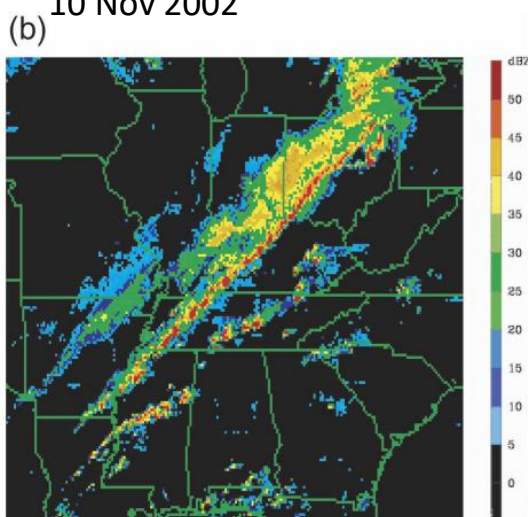
Note que a escala de velocidade w_s não pode ser aplicada aqui, pois a eficiência do entranhamento é diferente entre a turbulência induzida por cisalhamento e a turbulência convectiva.”

(1) Quais são as fontes de incerteza do modelo?

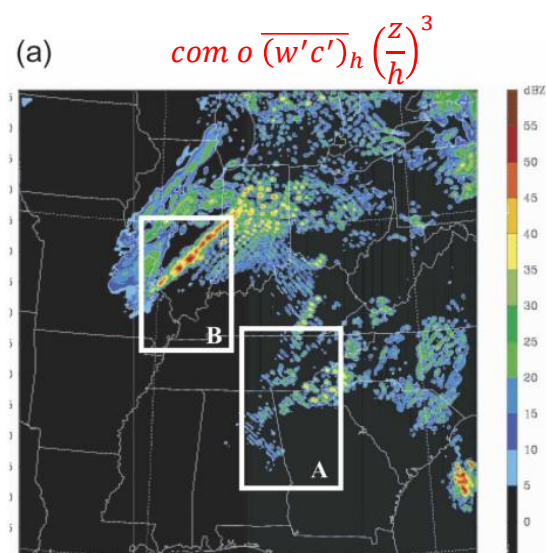
Processos físicos da turbulência na camada limite



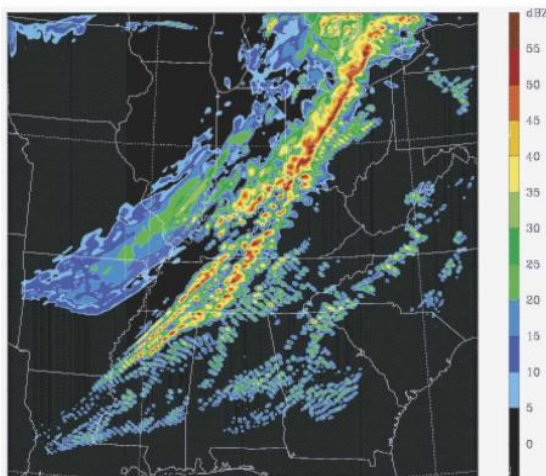
maximum (dBZ) at (a) 1800 UTC
10 Nov 2002



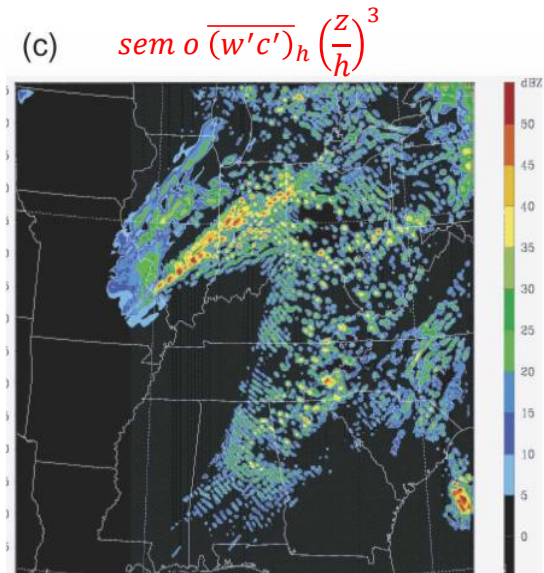
maximum (dBZ) at (a) 0000 UTC
11 Nov 2002



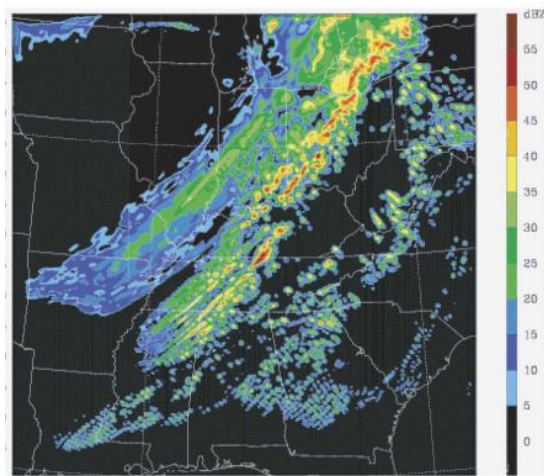
maximum (dBZ) at (a) 1800 UTC
(b) 10 Nov 2002



maximum (dBZ) at (a) 0000 UTC
11 Nov 2002



maximum (dBZ) at (a) 1800 UTC
(d) 10 Nov 2002



maximum (dBZ) at (a) 0000 UTC
11 Nov 2002



(1) Quais são as fontes de incerteza do modelo?

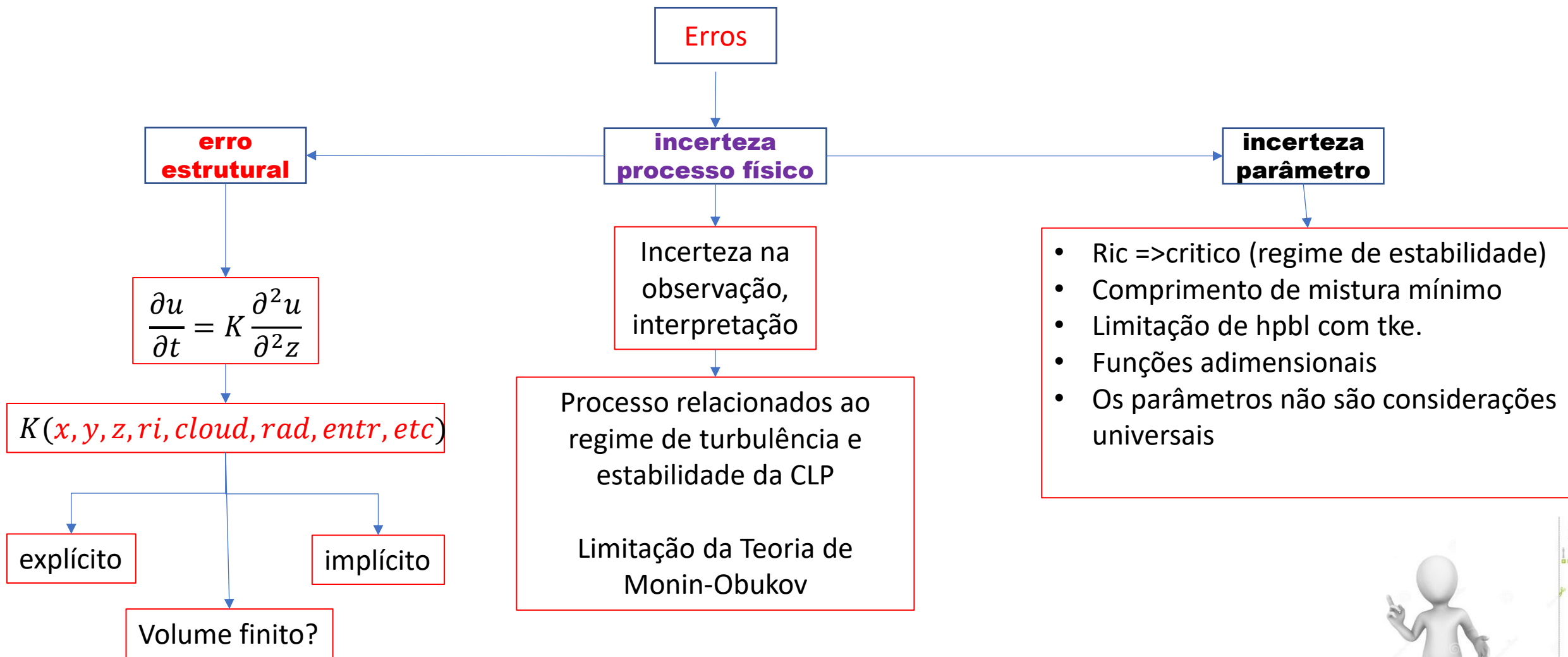
Processos físicos da turbulência na camada limite

$\overline{(w'c')}_h$ é o fluxo na camada de inversão

1. "O esquema YSU de (PBL) **aumenta a mistura na camada limite durante regimes de convecção livre induzida termicamente e a diminui a mistura durante regimes de convecção forçada induzida mecanicamente.**
2. A **mistura excessiva na camada de mistura na presença de ventos fortes é resolvida.**
3. O **crescimento excessivamente rápido da PBL no caso de Hong e Pan também é corrigido.**
4. .Consequentemente, **o novo esquema reproduz melhor a inibição convectiva.**
5. Isso ocorre porque a **camada limite do esquema YSU PBL permanece menos diluída pelo entranhamento, deixando mais energia para a convecção severa quando a frente a dispara."**

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite





Implementação da parametrização YSU BL

Implementation of a revised SBL scheme in the YSU BL package

“Propõem-se a adição da contribuição do fluxo de calor devido o entranhamento no topo da Camada Limite Planetária (PBL) ”

Noh et al 2003, modificaram o esquema original

$$\frac{\partial C}{\partial t} = \frac{\partial [-\overline{w'c'}]}{\partial z}$$

$$-\overline{w'c'} = K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) - \overline{(w'c')}_h \left(\frac{z}{h} \right)^n$$

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial z} \left[K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) \right]$$

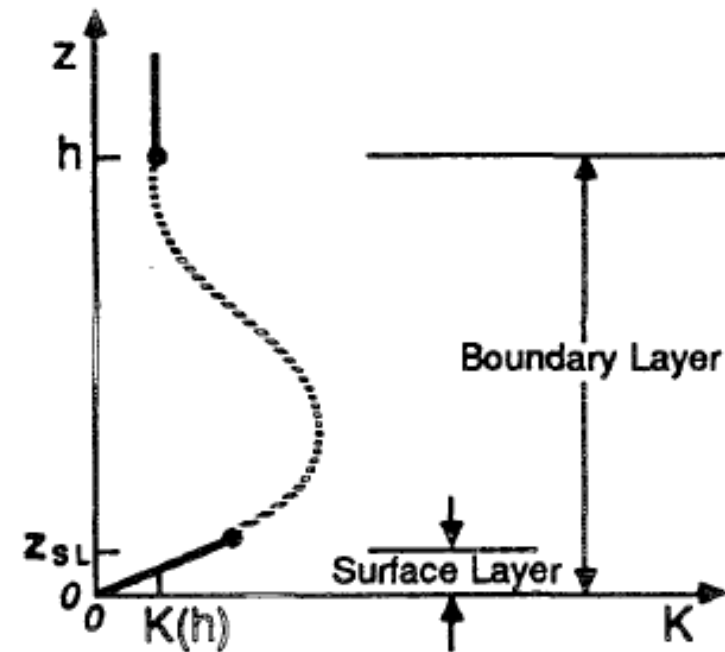


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Processos físicos da turbulência na camada limite

Implementation of a revised SBL scheme in the YSU BL package

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$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial z} \left\{ K_c \left(\frac{\partial C}{\partial z} - \gamma_c \right) - \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right\}$$

$$\frac{\partial C}{\partial t} = \left\{ \frac{\partial}{\partial z} K_c \frac{\partial C}{\partial z} - \frac{\partial}{\partial z} \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right) \right\}$$

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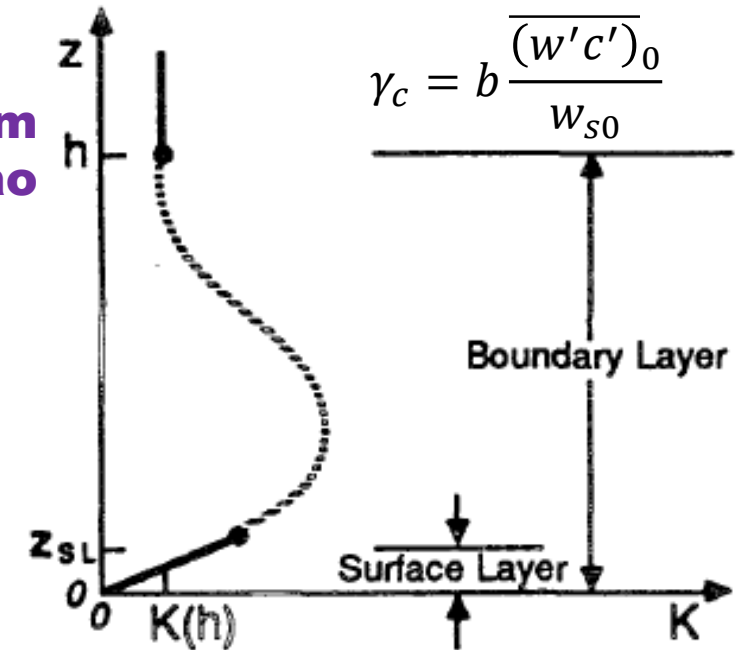


FIG. 1. Typical variation of eddy viscosity K with height in the boundary layer proposed by O'Brien (1970). Adopted from Stull (1988).

Implementation of a revised SBL scheme in the YSU BL package

$$\frac{\partial C}{\partial t} = \left\{ \frac{\partial}{\partial z} K_c \frac{\partial C}{\partial z} - \frac{\partial}{\partial z} \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right) \right\}$$

$$\int_k^{k+1} \frac{\partial C}{\partial t} dz = \left\{ \int_k^{k+1} \frac{\partial}{\partial z} K_c \frac{\partial C}{\partial z} dz - \int_k^{k+1} \frac{\partial}{\partial z} \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right) dz \right\}$$

$$\frac{\partial}{\partial t} \int_k^{k+1} C dz = \left\{ K_c \frac{\partial C}{\partial z} \Big|_k^{k+1} - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right) \Big|_k^{k+1} \right\}$$

$$\frac{\partial \bar{C}}{\partial t} = \left\{ \left(K_c \frac{\partial C}{\partial z} \right)^{k+1} - \left(K_c \frac{\partial C}{\partial z} \right)^k - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k \right\}$$

$$\frac{\partial \bar{C}}{\partial t} - \left(K_c \frac{\partial C}{\partial z} \right)^{k+1} + \left(K_c \frac{\partial C}{\partial z} \right)^k = \left\{ - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k \right\}$$

Implementation of a revised SBL scheme in the YSU BL package

$$\frac{\partial \bar{c}}{\partial t} - \left(K_c \frac{\partial \bar{c}}{\partial z} \right)^{k+1} + \left(K_c \frac{\partial \bar{c}}{\partial z} \right)^k = - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$\frac{C_k^{n+1} - C_k^{n-1}}{2\Delta t} - \left(K_{c,k+1}^n \frac{C_{k+1}^{n+1} - C_k^{n+1}}{\Delta z} \right)^{k+1} + \left(K_{c,k}^n \frac{C_k^{n+1} - C_{k-1}^{n+1}}{\Delta z} \right)^k = - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$\frac{C_k^{n+1} - C_k^{n-1}}{2\Delta t} - K_{c,k+1}^n \frac{C_{k+1}^{n+1} - C_k^{n+1}}{\Delta z} + K_{c,k}^n \frac{C_k^{n+1} - C_{k-1}^{n+1}}{\Delta z} = - \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$C_k^{n+1} - C_k^{n-1} - \frac{K_{c,k+1}^n 2\Delta t}{\Delta z} (C_{k+1}^{n+1} - C_k^{n+1}) + \frac{K_{c,k}^n 2\Delta t}{\Delta z} (C_k^{n+1} - C_{k-1}^{n+1}) = -2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$C_k^{n+1} - C_k^{n-1} + \left(-\frac{K_{c,k+1}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} + \frac{K_{c,k+1}^n 2\Delta t}{\Delta z} C_k^{n+1} \right) + \left(\frac{K_{c,k}^n 2\Delta t}{\Delta z} C_k^{n+1} - \frac{K_{c,k}^n 2\Delta t}{\Delta z} C_{k-1}^{n+1} \right) = -2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

$$-\frac{K_{c,k+1}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} + \frac{K_{c,k+1}^n 2\Delta t}{\Delta z} C_k^{n+1} + \frac{K_{c,k}^n 2\Delta t}{\Delta z} C_k^{n+1} + C_k^{n+1} + \left(-\frac{K_{c,k}^n 2\Delta t}{\Delta z} C_{k-1}^{n+1} \right) = C_k^{n-1} - 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h} \right)^3 \right)^k$$

(1)Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite



Implementation of a revised SBL scheme in the YSU BL package

$$-\frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} + \frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_k^{n+1} + \frac{K_{c_k}^n 2\Delta t}{\Delta z} C_k^{n+1} + C_k^{n+1} + \left(-\frac{K_{c_k}^n 2\Delta t}{\Delta z} C_{k-1}^{n+1}\right) = C_k^{n-1} - 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^k$$

$$-\frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} + \frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_k^{n+1} + \frac{K_{c_k}^n 2\Delta t}{\Delta z} C_k^{n+1} + C_k^{n+1} + \left(-\frac{K_{c_k}^n 2\Delta t}{\Delta z} C_{k-1}^{n+1}\right) = C_k^{n-1} - 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^k$$

$$\left(-\frac{K_{c_k}^n 2\Delta t}{\Delta z}\right) C_{k-1}^{n+1} + \left(\frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} + \frac{K_{c_k}^n 2\Delta t}{\Delta z} + 1\right) C_k^{n+1} - \frac{K_{c_{k+1}}^n 2\Delta t}{\Delta z} C_{k+1}^{n+1} = C_k^{n-1} - 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^{k+1} + 2\Delta t \left(K_c \gamma_c + \overline{(w'c')}_h \left(\frac{z}{h}\right)^3\right)^k$$

$$(AA)C_{k-1}^{n+1} + BB C_k^{n+1} - DD C_{k+1}^{n+1} = C_k^{n-1} + f1_k - f1_{k+1}$$

$$(AA)C_{k-1}^{n+1} + BB C_k^{n+1} - DD C_{k+1}^{n+1} = FF_k^{n-1}$$

$$\begin{bmatrix} BB & DD & 0 & 0 & 0 & 0 \\ AA & BB & DD & 0 & 0 & 0 \\ 0 & AA & BB & DD & 0 & 0 \\ 0 & 0 & AA & BB & DD & 0 \\ 0 & 0 & 0 & AA & BB & DD \\ 0 & 0 & 0 & 0 & AA & BB \end{bmatrix} * \begin{bmatrix} C_1^{n+1} \\ C_2^{n+1} \\ C_3^{n+1} \\ C_4^{n+1} \\ C_5^{n+1} \\ C_6^{n+1} \end{bmatrix} = \begin{bmatrix} FF_1^{n-1} \\ FF_2^{n-1} \\ FF_3^{n-1} \\ FF_4^{n-1} \\ FF_5^{n-1} \\ FF_6^{n-1} \end{bmatrix} \leftarrow \frac{\partial C}{\partial t} = \frac{\partial [-\overline{w'c'}]}{\partial z}$$

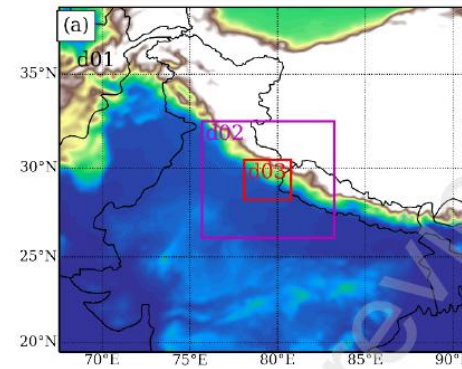


Alguns resultados interessantes



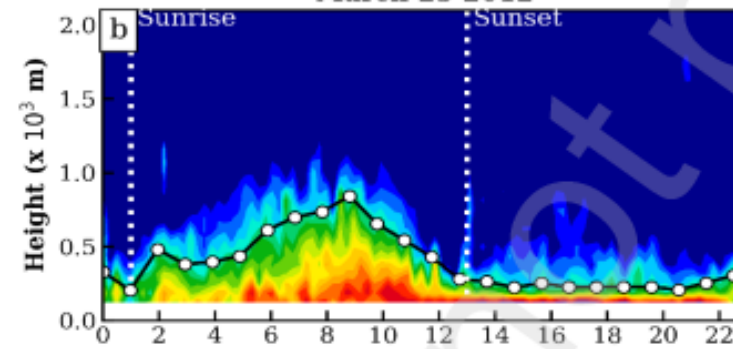
(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

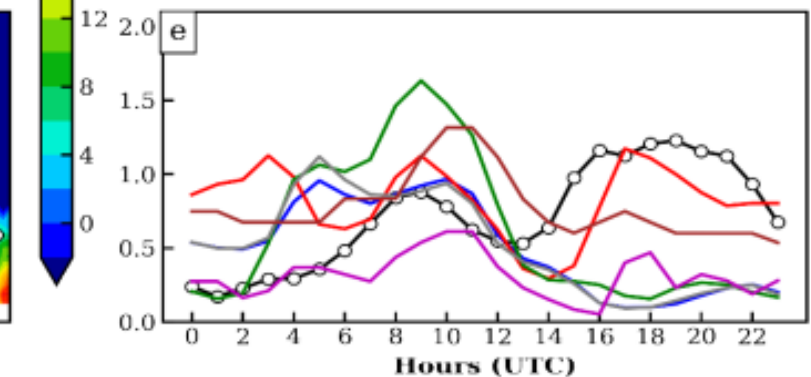
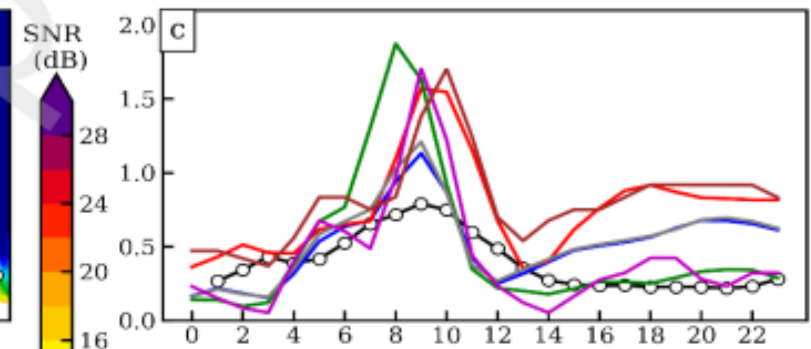
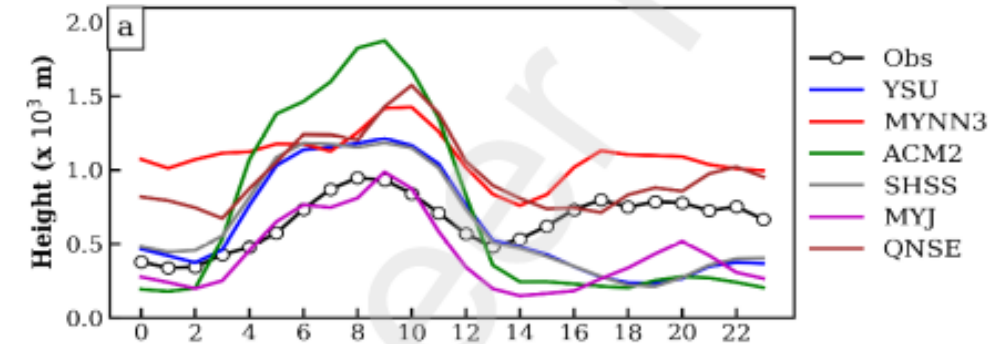
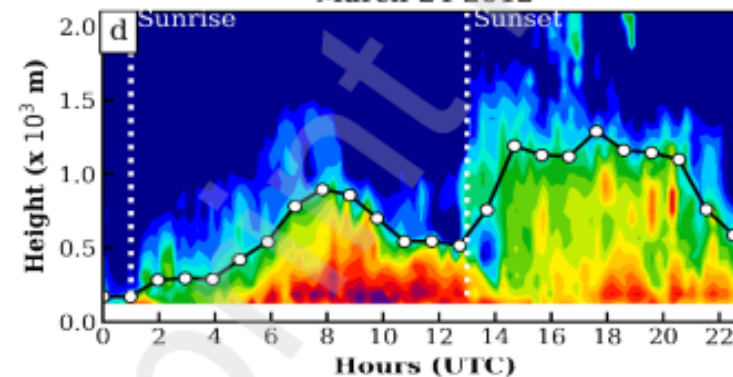


Himalaia

March 23 2012



March 24 2012



"Variação diurna da altura da (PBLH) (a) simulada usando diferentes esquemas de PBL e observações de RWP.

A variação da relação sinal-ruído (SNR) junto com a PBLH para

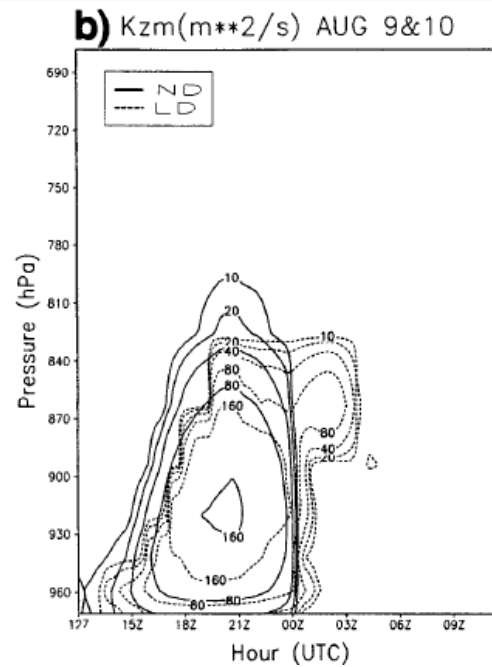
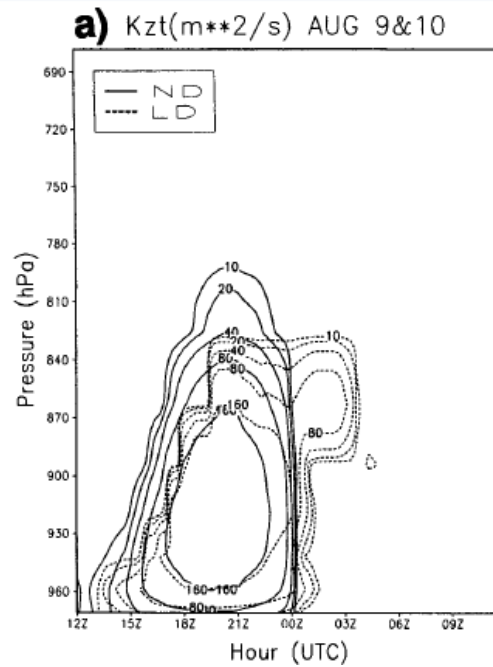
(b) 23 de março de 2012, com baixa PBLH durante a noite.

(d) 24 de março de 2012, com PBLH mais alta durante a noite."

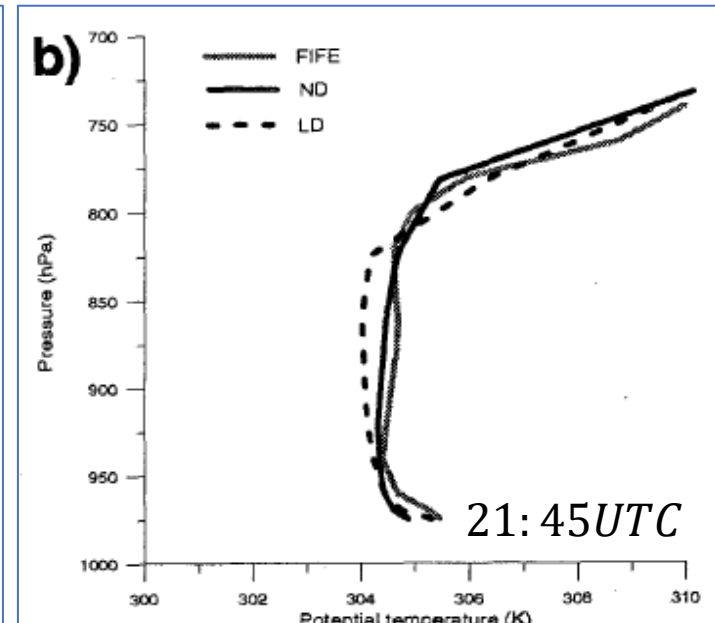
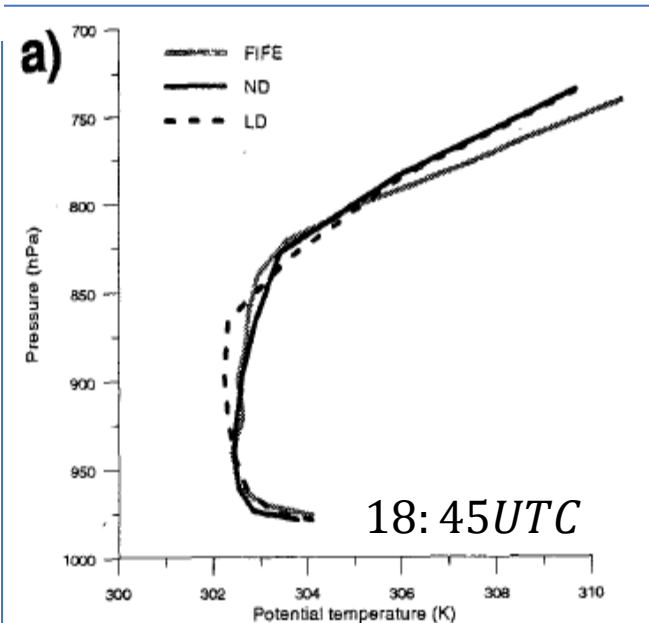
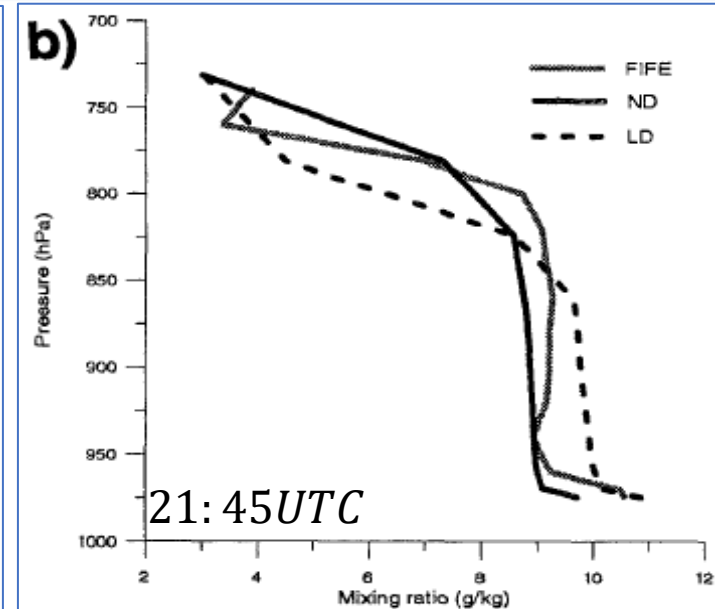
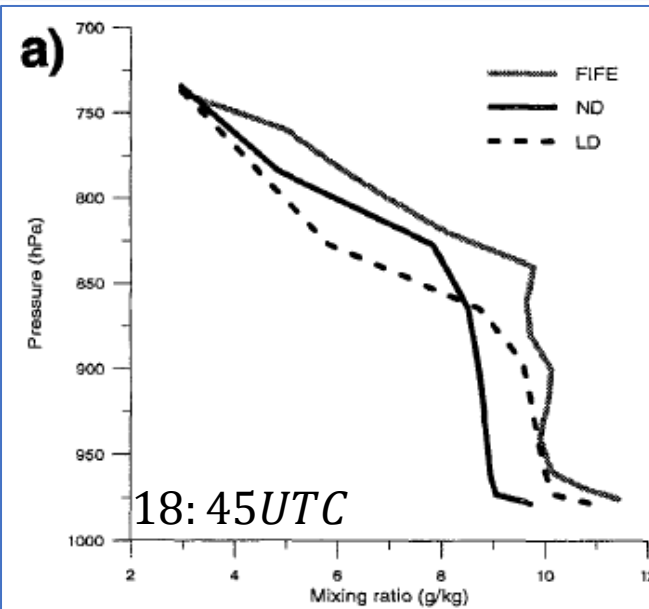


(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite



Perfil de difusividade de Eddies ($\frac{m^2}{s}$) linha pontilhada (local) e linhas contínuas não local



(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite

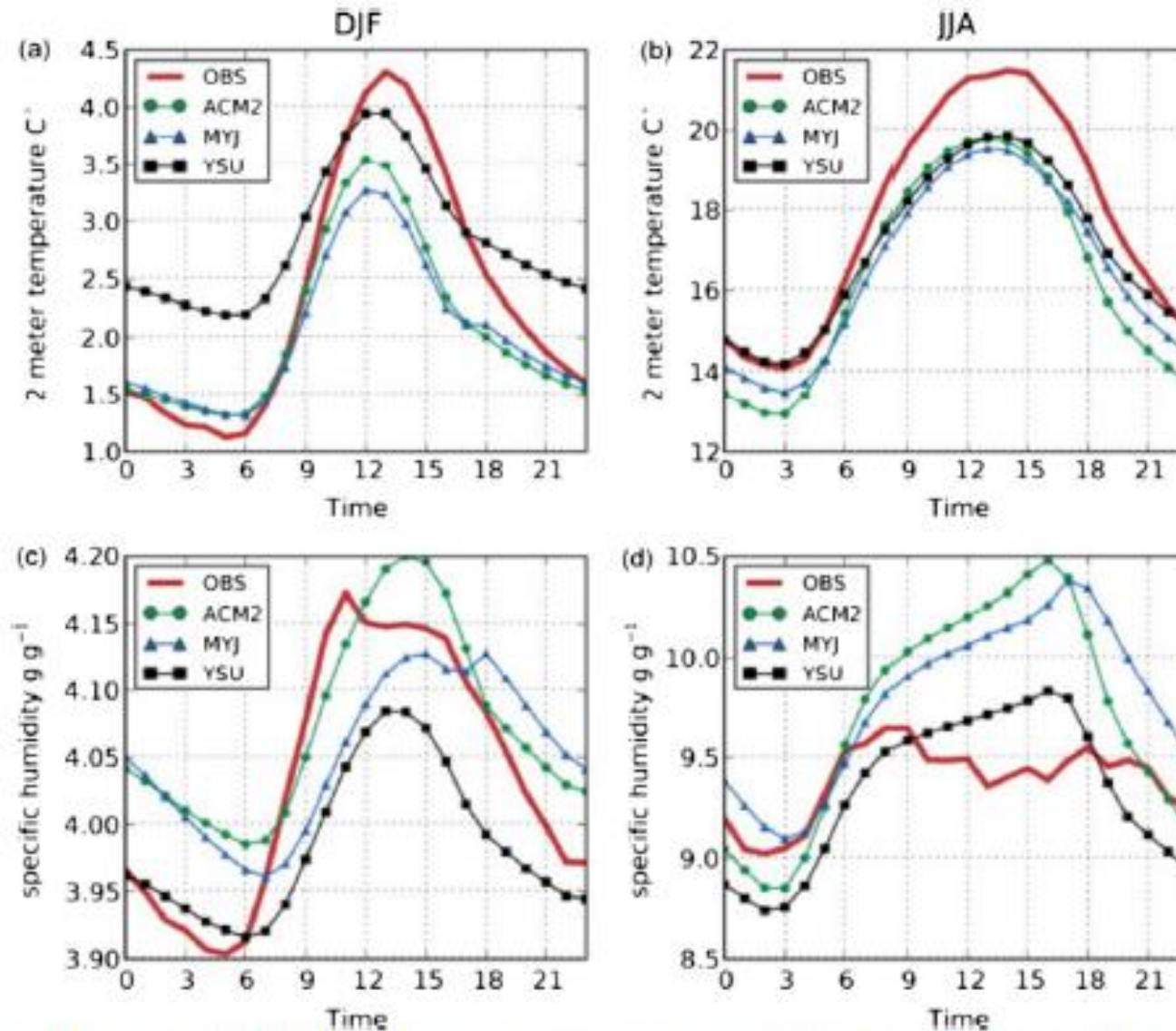
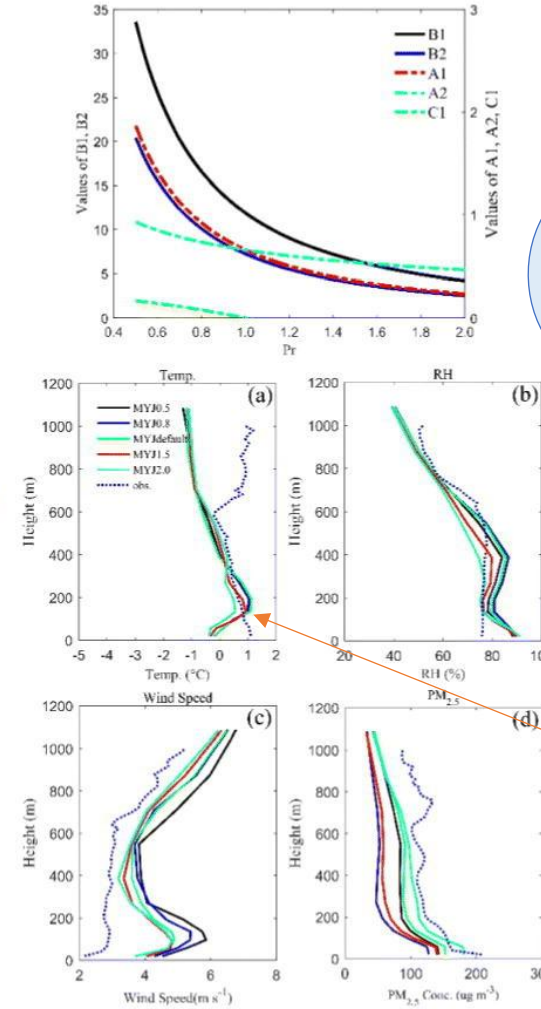
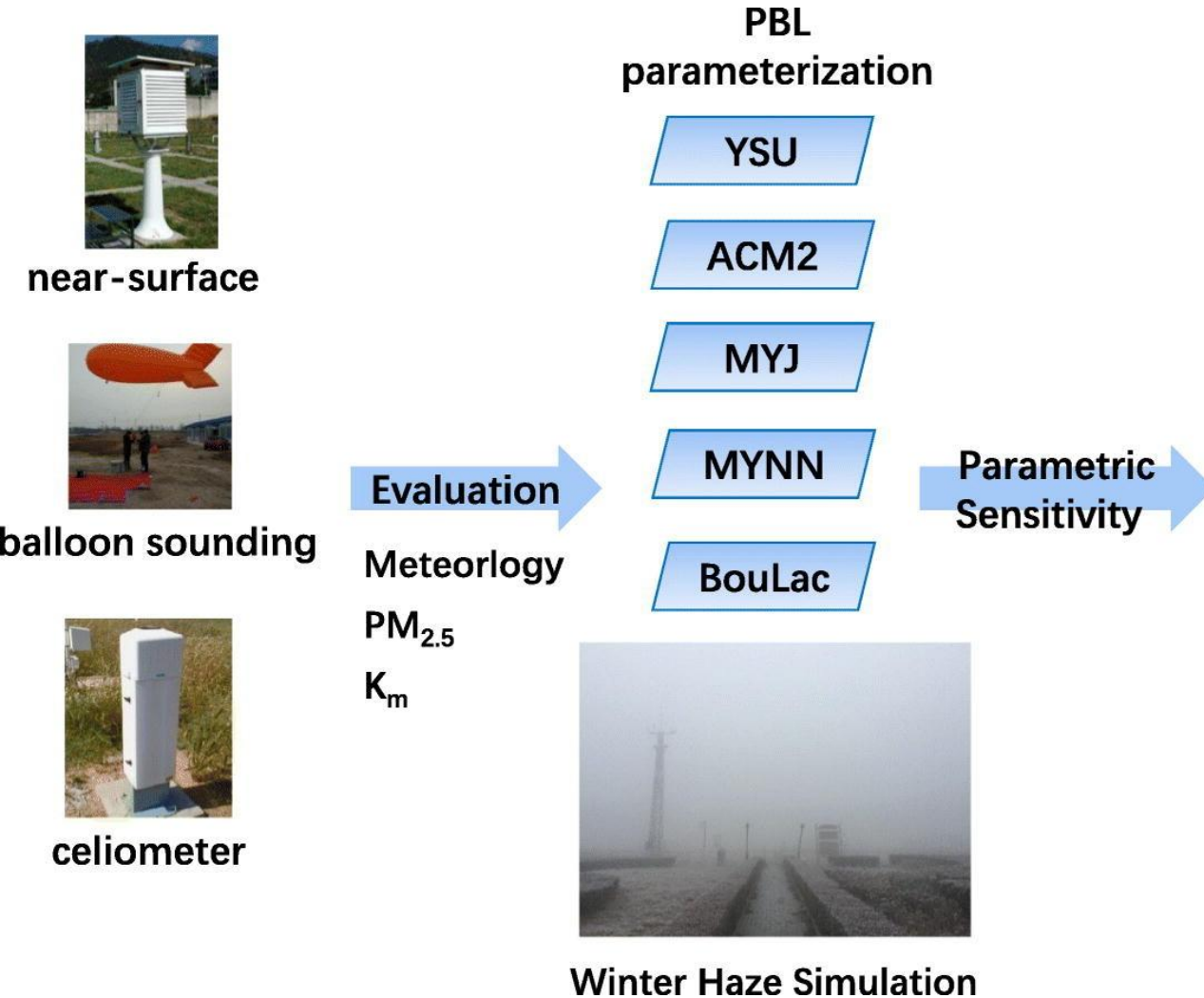


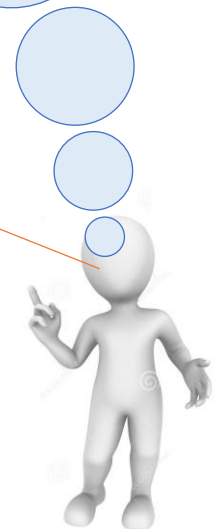
FIG. 3. Comparisons of WRF PBL schemes with European synoptic observations for temperatures during the (a) winter and (b) summer and for specific humidity during the (c) winter and (d) summer [from García-Díez et al. (2013)].

(1) Quais são as fontes de incerteza do modelo?

Processos físicos da turbulência na camada limite



A formulação do cálculo de K não incorpora corretamente os processos físicos. Os parâmetros podem estar mal calibrados K(x, y, z, ri, cloud, rad, entr, .)





Implementação da parametrização de Mellor Yamada Nino Nakanishi com fluxo de Massa

$$\overline{w'\phi'} = -K \frac{\partial \phi}{\partial z} + M_{\phi,u}(\phi_u - \phi) - M_{\phi,d}(\phi_d - \phi)$$

$$K_\phi = c_\phi L_k \sqrt{\bar{e}}$$

$$K_\phi = S_\phi L_k \sqrt{2\bar{e}}$$



Linearização da Equações de Navier Stokes

5.2 TURBULENT KINETIC ENERGY

A equação de Navier Stokes: conservação de momentum

$$\begin{matrix} i=1,2,3 \\ j=1,2,3 \\ K=1,2,3 \end{matrix} \quad \frac{Du_i}{Dt} = \frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial P}{\partial x_i} - \frac{\rho}{\rho_0} g \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j u_k + \nu \left(\frac{\partial^2 u_i}{\partial x_j^2} \right)$$

$$u_i = \bar{u}_i + u_i'$$

Aplique a Média de Reynolds na Variáveis

$$\frac{\partial(\bar{u}_i + u_i')}{\partial t} + (\bar{u}_j + u_j') \frac{\partial(\bar{u}_i + u_i')}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial(\bar{P} + P')}{\partial x_i} - g \frac{(\rho + \rho')}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k + u_k') + \nu \left(\frac{\partial^2(\bar{u}_i + u_i')}{\partial x_j^2} \right)$$

Expanda os termos

$$\begin{aligned} & \frac{\partial(\bar{u}_i)}{\partial t} + \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(u_i')}{\partial x_j} = \\ & -\frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - \frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$



A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} \frac{\partial(\bar{u}_i)}{\partial t} + \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(u_i')}{\partial x_j} = \\ - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - \frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$

Separa os termos na equação acima

$$\begin{aligned} \frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(u_i')}{\partial x_j} = \\ - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} \end{aligned}$$



Linearização da Equações de Navier Stokes

A equação de Navier Stokes: conservação de momentum

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + \boxed{(u_j') \frac{\partial(u_i')}{\partial x_j}} =$$
$$-\frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + \boxed{\frac{\partial(u_j' u_i')}{\partial x_j} - (u_i') \frac{\partial(u_j')}{\partial x_j}}$$
$$= -\frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

Aplique as media de Reynolds

$$\frac{\partial(\bar{\bar{u}}_i)}{\partial t} + (\bar{\bar{u}}_j) \frac{\partial(\bar{\bar{u}}_i)}{\partial x_j} + \frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \overline{(u_i')} \frac{\partial(\overline{u_j'})}{\partial x_j} + (\overline{u_j'}) \frac{\partial(\bar{\bar{u}}_i)}{\partial x_j}$$
$$= -\frac{1}{\rho_0} \frac{\partial(\bar{\bar{P}})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{\bar{u}}_k) + \nu \frac{\partial^2(\bar{\bar{u}}_i)}{\partial x_j^2}$$



Linearização da Equações de Navier Stokes



A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} \frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + \frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \overline{(u_i')} \frac{\partial(\overline{u_j'})}{\partial x_j} + (\overline{u_j'}) \frac{\partial(\bar{u}_i)}{\partial x_j} \\ = -\frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} \end{aligned}$$

Aplique as considerações da media de Reynolds

$$\bar{\bar{u}}_i = \bar{u}_i$$

$$\overline{u_j' u_i'} \neq 0$$

$$\overline{u_j'} = 0$$

$$\overline{u_i'} = 0$$

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} = -\frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$



- **Equação primitiva não linear para o escoamento da Atmosfera**

$$\frac{Du_i}{Dt} = \frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial P}{\partial x_i} - \frac{\rho}{\rho_0} g \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j u_k + \nu \left(\frac{\partial^2 u_i}{\partial x_j^2} \right)$$

- **Equação Governante Linearizada do Estado Médio do escoamento da Atmosfera**

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} = -\frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$



Linearização da Equações de Navier Stokes



5.2 TURBULENT KINETIC ENERGY A equação de Navier Stokes: conservação de momentum

$$\frac{\partial(\bar{u}_i)}{\partial t} + \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + (u_j') \frac{\partial(\bar{u}_i)}{\partial x_j} + (u_j') \frac{\partial(u_i')}{\partial x_j} =$$

$$-\frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - \frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2} + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$

Separe os termos com perturbação que se cancelariam com a media de Reynolds

$$\frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \boxed{(u_j') \frac{\partial(\bar{u}_i)}{\partial x_j}} + \boxed{(u_j') \frac{\partial(u_i')}{\partial x_j}}$$

$$= -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$

$$\frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \boxed{\frac{\partial(u_j' \bar{u}_i)}{\partial x_j} - (\bar{u}_i) \frac{\partial(u_j')}{\partial x_j}} + \boxed{\frac{\partial(u_j' u_i')}{\partial x_j} - (u_i') \frac{\partial(u_j')}{\partial x_j}}$$

$$= -\frac{1}{\rho} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$



Linearização da Equações de Navier Stokes

5.2 TURBULENT KINETIC ENERGY A equação de Navier Stokes: conservação de momentum

$$\frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \boxed{(u_j') \frac{\partial(\bar{u}_i)}{\partial x_j}} + \boxed{(u_j') \frac{\partial(u_i')}{\partial x_j}} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$

Aplique a derivada do produto nos termos em destaque:

$$\begin{aligned} \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \boxed{\frac{\partial(u_j' \bar{u}_i)}{\partial x_j} - (\bar{u}_i) \frac{\partial(u_j')}{\partial x_j}} + \boxed{\frac{\partial(u_j' u_i')}{\partial x_j} - (u_i') \frac{\partial(u_j')}{\partial x_j}} \\ = -\frac{1}{\rho} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$



Linearização da Equações de Navier Stokes



5.2 TURBULENT KINETIC ENERGY A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} \frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \frac{\partial(u_j' \bar{u}_i)}{\partial x_j} - (\bar{u}_i) \frac{\partial(u_j')}{\partial x_j} + \frac{\partial(u_j' u_i')}{\partial x_j} - (u_i') \frac{\partial(u_j')}{\partial x_j} \\ = -\frac{1}{\rho} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2} \end{aligned}$$

Considere que a propriedade da equação da continuidade para o fluxo $\nabla \cdot \vec{V} = 0$

$$\nabla \cdot \vec{V}' = \frac{\partial(u_j')}{\partial x_j} = 0, \quad j = 1, 2, 3$$

$$\frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \frac{\partial(u_j' \bar{u}_i)}{\partial x_j} + \frac{\partial(u_j' u_i')}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$



Linearização da Equações de Navier Stokes



5.2 TURBULENT KINETIC ENERGY A equação de Navier Stokes: conservação de momentum

$$\frac{\partial(u_i')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i')}{\partial x_j} + \frac{\partial(u_j' \bar{u}_i)}{\partial x_j} + \frac{\partial(u_j' u_i')}{\partial x_j} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_i} - g \frac{\rho'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k') + \nu \frac{\partial^2(u_i')}{\partial x_j^2}$$



**Gorge_mellor_Analytic Prediction of the
Properties of Stratified Planetary Surface
Layers_atsc-1520-0469_1973**



1.1 A Teoria de Similaridade de Monin-Obukhov



As equações são apresentadas de uma forma geral considerável para que, em princípio, possam ser integradas para simular, por exemplo, uma camada limite planetária completa.

Aqui, no entanto, restringimos a atenção à região da superfície de fluxo constante, evitando assim, por enquanto, o considerável esforço computacional necessário para a camada completa.

No entanto, este é um primeiro passo lógico, uma vez que é possível comparar diretamente com os dados de fluxo constante de Businger et al. (1971) na forma de variáveis de similaridade de Monin-Obukhoff.



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

As equações de movimento para a velocidade média $\bar{u}_j$; e a temperatura potencial média $\bar{\theta}$, As barras superiores representam as médias do conjunto e os termos minúsculos, u'_k e θ' , são os componentes flutuantes da velocidade e da temperatura e são governados por

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial \bar{u}_j}{\partial t} + \bar{u}_k \frac{\partial (\bar{u}_j)}{\partial x_k} + \bar{u}_j \left[\frac{\partial (\bar{u}_k)}{\partial x_k} \right] + \frac{\partial (\overline{u'_k u'_j})}{\partial x_k} + \epsilon_{j,k,l} f_k \bar{u}_l = -\frac{\partial P}{\partial x_j} - g_j \beta \bar{\theta} + \nu \nabla^2 \bar{u}_j \quad (2)$$

$$\frac{\partial \bar{u}_j}{\partial t} + \frac{\partial (\bar{u}_k \bar{u}_j)}{\partial x_k} + \frac{\partial (\overline{u'_k u'_j})}{\partial x_k} + \epsilon_{j,k,l} f_k \bar{u}_l = -\frac{\partial P}{\partial x_j} - g_j \beta \bar{\theta} + \nu \nabla^2 \bar{u}_j \quad (2)$$

$$\frac{\partial \bar{u}_j}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k \bar{u}_j + \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k \bar{u}_l = -\frac{\partial P}{\partial x_j} - g_j \beta \bar{\theta} + \nu \nabla^2 \bar{u}_j \quad (2)$$

$$\frac{\partial \bar{\theta}}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k \bar{\theta} + \overline{u'_k \theta'}) = \alpha \nabla^2 \bar{\theta} \quad (3)$$

onde P é a pressão cinemática média, $g_j = (0, 0, -g)$ vetor de gravidade, $f_j = (0, f_y, f)$ o parâmetro de Coriolis (o componente vertical de não terá subscrito), $\beta = \left(\frac{\partial \rho}{\partial T} \right)_p / \rho$ o coeficiente de expansão térmica, ν é a viscosidade cinemática e α é a condutividade térmica cinemática (ou difusividade térmica).



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\frac{\partial u'_i}{\partial x_i} = 0 \quad (4)$$

$$\frac{\partial(u'_j)}{\partial t} + (\overline{u_k}) \frac{\partial(u'_j)}{\partial x_k} + \frac{\partial(u'_k \overline{u_j})}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_j} - g \frac{\rho'}{\rho_0} \delta_{j3} - 2\Omega \varepsilon_{jki} \eta_k (u'_k) + \nu \frac{\partial^2(u'_j)}{\partial x_k^2}$$

Se você aplicar a media de media de Reynolds???? Os termos turbulentos são nulos $\overline{u'_j} = 0$

$$\frac{\partial(\overline{u'_j})}{\partial t} + (\overline{\overline{u_k}}) \frac{\partial(\overline{u'_j})}{\partial x_k} + \frac{\partial(\overline{u'_k \overline{u_j}})}{\partial x_k} + \frac{\partial(\overline{u'_k u'_j})}{\partial x_k} = -\frac{1}{\rho_0} \frac{\partial(\overline{P'})}{\partial x_j} - g \frac{\overline{\rho'}}{\rho_0} \delta_{j3} - 2\Omega \varepsilon_{jki} \eta_k (\overline{u'_k}) + \nu \frac{\partial^2(\overline{u'_j})}{\partial x_k^2}$$

Se você aplicar a media de media de Reynolds???? então $\overline{u'_j} = 0$

$$\frac{\partial(\overline{u'_k u'_j})}{\partial x_k} = \frac{\partial(\overline{u'_k u'_j})}{\partial x_k} = 0$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\frac{\partial u'_i}{\partial x_i} = 0 \quad (4)$$

$$\frac{\partial(u'_j)}{\partial t} + (\overline{u_k}) \frac{\partial(u'_j)}{\partial x_k} + \frac{\partial(u'_k \overline{u_j})}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_j} - g \frac{\rho'}{\rho_0} \delta_{j3} - 2\Omega \varepsilon_{jki} \eta_k (u'_k) + \nu \frac{\partial^2(u'_j)}{\partial x_k^2}$$

Se você aplicar a media de media de Reynolds???? Os termos turbulentos são nulos $\overline{u'_j} = 0$

$$\frac{\partial(\overline{u'_k u'_j})}{\partial x_k} = \frac{\partial(\overline{u'_k u'_j})}{\partial x_k} = 0$$

$$u'_j \frac{\partial \overline{u_k}}{\partial x_k} = 0$$

$$\frac{\partial u'_j}{\partial t} + \left(\overline{u_k} \frac{\partial u'_j}{\partial x_k} + u'_j \frac{\partial \overline{u_k}}{\partial x_k} + \frac{\partial(\overline{u_j u'_k})}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} - \frac{\partial(\overline{u'_k u'_j})}{\partial x_k} \right) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

Como as equações médias (1), (2) e (3) envolvem a tensão de Reynolds $\overline{u'_i u'_j}$ e os momentos de condução de calor, $u'_i \theta'$, obtemos suas equações governantes de (5) e (6) usando a (4)



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\frac{\partial u'_i}{\partial x_i} = 0 \quad (4)$$

$$\frac{\partial(u'_j)}{\partial t} + (\overline{u_k}) \frac{\partial(u'_j)}{\partial x_k} + \frac{\partial(\overline{u'_k u_j})}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} = -\frac{1}{\rho_0} \frac{\partial(P')}{\partial x_j} - g \frac{\rho'}{\rho_0} \delta_{j3} - 2\Omega \varepsilon_{jki} \eta_k(u'_k) + \nu \frac{\partial^2(u'_j)}{\partial x_k^2}$$

$$\frac{\partial u'_j}{\partial t} + \left(\overline{u_k} \frac{\partial u'_j}{\partial x_k} + u'_j \frac{\partial \overline{u_k}}{\partial x_k} + \frac{\partial(\overline{u_j u'_k})}{\partial x_k} + \frac{\partial(u'_k u'_j)}{\partial x_k} - \frac{\partial(\overline{u'_k u'_j})}{\partial x_k} \right) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j$$

$$\frac{\partial u'_j}{\partial t} + \left(\frac{\partial \overline{u_k} u'_j}{\partial x_k} + \frac{\partial \overline{u_j u'_k}}{\partial x_k} + \frac{\partial u'_k u'_j}{\partial x_k} - \frac{\partial \overline{u'_k u'_j}}{\partial x_k} \right) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

$$\frac{\partial u'_j}{\partial t} + \frac{\partial}{\partial x_k} (\overline{u_k} u'_j + \overline{u_j u'_k} + u'_k u'_j - \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

$$\frac{\partial \theta'}{\partial t} + \frac{\partial}{\partial x_k} (\overline{\theta} u'_k + \overline{u_k \theta'} + u'_k \theta' + \overline{u'_k \theta'}) = \alpha \nabla^2 \theta' \quad (6)$$

Como as equações médias (1), (2) e (3) envolvem a tensão de Reynolds $\overline{u'_i u'_j}$ e os momentos de condução de calor, $u'_i \theta'$, obtemos suas equações governantes de (5) e (6) usando a (4)



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\frac{\partial u'_j}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k u'_j + \bar{u}_j u'_k + u'_k u'_j - \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

$$u'_i \frac{\partial u'_j}{\partial t} + u'_i \frac{\partial}{\partial x_k} (\bar{u}_k u'_j + \bar{u}_j u'_k + u'_k u'_j - \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k u'_l u'_i = -u'_i \frac{\partial P'}{\partial x_j} - g_j \beta u'_i \theta' + \nu u'_i \nabla^2 u'_j \quad (5a)$$

$$u'_j \frac{\partial u'_i}{\partial t} + u'_j \frac{\partial}{\partial x_k} (\bar{u}_k u'_i + \bar{u}_i u'_k + u'_k u'_i - \overline{u'_k u'_i}) + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_j \frac{\partial P'}{\partial x_i} - g_i \beta u'_j \theta' + \nu u'_j \nabla^2 u'_i \quad (5b)$$

Mude i por j e j por i

Some a eq 5-a com 5-b

$$\begin{aligned} & \left(u'_i \frac{\partial u'_j}{\partial t} + u'_j \frac{\partial u'_i}{\partial t} \right) + \left(u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right) + \left(u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right) - \left(u'_i \frac{\partial \overline{u'_k u'_j}}{\partial x_k} + u'_j \frac{\partial \overline{u'_k u'_i}}{\partial x_k} \right) \\ & + \epsilon_{j,k,l} f_k u'_l u'_i + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_i \frac{\partial P'}{\partial x_j} - u'_j \frac{\partial P'}{\partial x_i} - g_j \beta u'_i \theta' - g_i \beta u'_j \theta' + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \quad (5c) \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(u'_i \frac{\partial u'_j}{\partial t} + u'_j \frac{\partial u'_i}{\partial t} \right) + \left(u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right) + \left(u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right) - \left(u'_i \frac{\partial \overline{u'_k u'_j}}{\partial x_k} + u'_j \frac{\partial \overline{u'_k u'_i}}{\partial x_k} \right) \\ & + \epsilon_{j,k,l} f_k u'_l u'_i + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_i \frac{\partial P'}{\partial x_j} - u'_j \frac{\partial P'}{\partial x_i} - g_j \beta u'_i \theta' - g_i \beta u'_j \theta' + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$

$$\begin{aligned} & \left(u'_i \frac{\partial u'_j}{\partial t} + u'_j \frac{\partial u'_i}{\partial t} \right) + \left(u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right) - \left(u'_i \frac{\partial \overline{u'_k u'_j}}{\partial x_k} + u'_j \frac{\partial \overline{u'_k u'_i}}{\partial x_k} \right) \\ & + \epsilon_{j,k,l} f_k u'_l u'_i + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_i \frac{\partial P'}{\partial x_j} - u'_j \frac{\partial P'}{\partial x_i} - g_j \beta u'_i \theta' - g_i \beta u'_j \theta' + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} = \frac{\partial \bar{u}_k}{\partial x_k} u'_i u'_j + \bar{u}_k \frac{\partial u'_i}{\partial x_k} u'_j + \bar{u}_k u'_i \frac{\partial u'_j}{\partial x_k} \Rightarrow \boxed{\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} - \bar{u}_k \frac{\partial u'_i u'_j}{\partial x_k}} = u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_i \bar{u}_k}{\partial x_k}$$

$$\frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} = \bar{u}_i \frac{\partial u'_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} + u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \Rightarrow \boxed{\frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} - \bar{u}_i \frac{\partial u'_j u'_k}{\partial x_k}} = u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} + u'_k \frac{\partial u'_j \bar{u}_i}{\partial x_k}$$

$$\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} = \bar{u}_j \frac{\partial u'_i u'_k}{\partial x_k} + u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} + u'_i \frac{\partial u'_k \bar{u}_j}{\partial x_k} \Rightarrow \boxed{\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} - \bar{u}_j \frac{\partial u'_i u'_k}{\partial x_k}} = u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} + u'_i \frac{\partial u'_k \bar{u}_j}{\partial x_k}$$

$$\frac{\partial u'_k u'_i u'_j}{\partial x_k} = u'_k \frac{\partial u'_i u'_j}{\partial x_k} + u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k} \Rightarrow \boxed{\frac{\partial u'_k u'_i u'_j}{\partial x_k} - u'_k \frac{\partial u'_i u'_j}{\partial x_k}} = u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k}$$

$$\frac{\partial u'_i \overline{u'_k u'_j}}{\partial x_k} = u'_i \frac{\partial \overline{u'_k u'_j}}{\partial x_k} + \overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \Rightarrow \frac{\partial u'_i \overline{u'_k u'_j}}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} = u'_i \frac{\partial \overline{u'_k u'_j}}{\partial x_k}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\left(u'_i \frac{\partial u'_j}{\partial t} + u'_j \frac{\partial u'_i}{\partial t} \right) + \left(u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right) + \left(u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right) - \left(u'_i \frac{\partial \bar{u}'_k u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}'_k u'_i}{\partial x_k} \right) \\ + \epsilon_{j,k,l} f_k u'_l u'_i + \epsilon_{i,k,l} f_k u'_l u'_j = -u'_i \frac{\partial P'}{\partial x_j} - u'_j \frac{\partial P'}{\partial x_i} - g_j \beta u'_i \theta' - g_i \beta u'_j \theta' + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i$$

Aplica-se as derivadas anteriores

$$\Rightarrow \frac{\partial (\bar{u}_k u'_j) u'_i}{\partial x_k} = u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + \bar{u}_k u'_j \frac{\partial u'_i}{\partial x_k} \quad \Rightarrow \frac{\partial (u'_k u'_j) u'_i}{\partial x_k} = u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} + u'_k u'_j \frac{\partial u'_i}{\partial x_k} \quad \Rightarrow \frac{\partial (\bar{u}_j u'_k) u'_i}{\partial x_k} = u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + \bar{u}_j u'_k \frac{\partial u'_i}{\partial x_k} \\ \Rightarrow \frac{\partial (\bar{u}_k u'_i) u'_j}{\partial x_k} = u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} + \bar{u}_k u'_i \frac{\partial u'_j}{\partial x_k} \quad \Rightarrow \frac{\partial (u'_k u'_i) u'_j}{\partial x_k} = u'_j \frac{\partial u'_k u'_i}{\partial x_k} + u'_k u'_i \frac{\partial u'_j}{\partial x_k} \quad \Rightarrow \frac{\partial (\bar{u}_i u'_k) u'_j}{\partial x_k} = u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} + \bar{u}_i u'_k \frac{\partial u'_j}{\partial x_k}$$

$$\left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} - \bar{u}_k u'_j \frac{\partial u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} - \bar{u}_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} - u'_k u'_j \frac{\partial u'_i}{\partial x_k} \right] + \left[\frac{\partial u'_k u'_j u'_i}{\partial x_k} - u'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) \\ + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} - \bar{u}_j u'_k \frac{\partial u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} - \bar{u}_i u'_k \frac{\partial u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \bar{u}'_k u'_i u'_j}{\partial x_k} - \bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}'_k u'_j u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) \\ + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} - P' \frac{\partial u'_i}{\partial x_j} \right] - \left[\frac{\partial u'_j P'}{\partial x_i} - P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i$$



1.1 A Teoria de Similaridade de Monin-Obukhov



$$\begin{aligned} & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} - \bar{u}_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_k \mathbf{u}'_j \mathbf{u}'_i}{\partial x_k} - \bar{u}_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} - u'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial u'_k \mathbf{u}'_j \mathbf{u}'_i}{\partial x_k} - u'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} - \bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} - \bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} - \bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i \end{aligned}$$

$$\begin{aligned} & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\bar{u}_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial u'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-u'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - u'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} \right] + \left[-\bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} - \bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i \end{aligned}$$

$$\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} = \mathbf{u}'_i \mathbf{u}'_j \frac{\partial \bar{u}_k}{\partial x_k} + \bar{u}_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} + \bar{u}_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \Rightarrow \frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} = \bar{u}_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} + \bar{u}_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



$$\begin{aligned} & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\bar{u}_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\mathbf{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \mathbf{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} \right] + \left[-\bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} - \bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i \end{aligned}$$

$$\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} = \mathbf{u}'_i \mathbf{u}'_j \frac{\partial \bar{u}_k}{\partial x_k} + \bar{u}_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} + \bar{u}_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \Rightarrow \frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} = \bar{u}_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} + \bar{u}_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k}$$

$$\begin{aligned} & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} \right] + \left[-\bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} - \bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i \end{aligned}$$



$$\begin{aligned}
& \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) \\
& + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} \right] + \left[-\bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} - \bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\
& + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i
\end{aligned}$$

$$\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} = \mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} + \bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k} + \bar{u}_j \mathbf{u}'_i \frac{\partial \mathbf{u}'_k}{\partial x_k}$$

$$\frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} = \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} + \bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k} + \bar{u}_i \mathbf{u}'_j \frac{\partial \mathbf{u}'_k}{\partial x_k}$$

$$\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} - \mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} = +\bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k} + \bar{u}_j \mathbf{u}'_i \frac{\partial \mathbf{u}'_k}{\partial x_k}$$

$$\frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} - \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} = +\bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k} + \bar{u}_i \mathbf{u}'_j \frac{\partial \mathbf{u}'_k}{\partial x_k}$$

$$\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} - \mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} = \bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k}$$

$$\frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} - \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} = +\bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k}$$



$$\begin{aligned}
& \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[- \frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[- \frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) \\
& + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} \right] + \left[- \bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} - \bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\
& + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i
\end{aligned}$$

$$\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} - \mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} = \bar{u}_j \mathbf{u}'_k \frac{\partial \mathbf{u}'_i}{\partial x_k}$$

$$\frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} - \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} = + \bar{u}_i \mathbf{u}'_k \frac{\partial \mathbf{u}'_j}{\partial x_k}$$

$$\begin{aligned}
& \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[- \frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[- \frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) \\
& + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} \right] + \left[- \frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} - \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} + \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) \\
& - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} - \bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) \\
& = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i
\end{aligned}$$



$$\begin{aligned} & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} \right] + \left[-\frac{\partial \bar{u}_j \mathbf{u}'_i \mathbf{u}'_k}{\partial x_k} + \mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} - \frac{\partial \bar{u}_i \mathbf{u}'_j \mathbf{u}'_k}{\partial x_k} + \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) \\ & - \left(\left[\frac{\partial \mathbf{u}'_i \mathbf{u}'_k \mathbf{u}'_j}{\partial x_k} - \overline{\mathbf{u}'_k \mathbf{u}'_j} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \mathbf{u}'_k \mathbf{u}'_i}{\partial x_k} - \overline{\mathbf{u}'_k \mathbf{u}'_i} \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) \\ & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i \end{aligned}$$

$$\begin{aligned} & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] + \left[-\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} + \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) \\ & - \left(\left[\frac{\partial \mathbf{u}'_i \mathbf{u}'_k \mathbf{u}'_j}{\partial x_k} - \overline{\mathbf{u}'_k \mathbf{u}'_j} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \mathbf{u}'_k \mathbf{u}'_i}{\partial x_k} - \overline{\mathbf{u}'_k \mathbf{u}'_i} \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) \\ & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 \mathbf{u}'_j + \nu \mathbf{u}'_j \nabla^2 \mathbf{u}'_i \end{aligned}$$



$$\begin{aligned} & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\mathbf{u}'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + \mathbf{u}'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \overline{u'_k u'_j}}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \overline{u'_k u'_i}}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} - P' \frac{\partial \mathbf{u}'_i}{\partial x_j} \right] - \left[\frac{\partial \mathbf{u}'_j P'}{\partial x_i} - P' \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \mathbf{u}'_i \nabla^2 u'_j + \nu \mathbf{u}'_j \nabla^2 u'_i \end{aligned}$$

$$\begin{aligned} & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\mathbf{u}'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + \mathbf{u}'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \overline{u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \overline{u'_k u'_i}}{\partial x_k} \right] \right) \\ & - \left(\left[-\overline{u'_k u'_j} \frac{\partial \mathbf{u}'_i}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) \\ & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right) - 2\nu \left(\frac{\partial \mathbf{u}'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right) \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\ & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\ & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial u'_i u'_j}{\partial x_k} \right) - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right) \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \overline{u_j}}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \overline{u_i}}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \overline{u'_i u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial \overline{u'_j u'_k u'_i}}{\partial x_k} \right] \right) \\
 & - \left(\left[-\overline{u'_k u'_j} \frac{\partial \overline{u'_i}}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial \overline{u'_j}}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\
 & = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \overline{P'} \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \overline{u_j}}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \overline{u_i}}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \overline{u'_i u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial \overline{u'_j u'_k u'_i}}{\partial x_k} \right] \right) \\
 & + \left(\left[\overline{u'_k u'_j} \frac{\partial \overline{u'_i}}{\partial x_k} + \overline{u'_k u'_i} \frac{\partial \overline{u'_j}}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\
 & = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \overline{P'} \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \overline{u_j}}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \overline{u_i}}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \overline{u'_i u'_k u'_j}}{\partial x_k} \right] + \left[\frac{\partial \overline{u'_j u'_k u'_i}}{\partial x_k} \right] \right) \\
 & + \left(\left[\overline{u'_k u'_j} \frac{\partial \overline{u'_i}}{\partial x_k} + \overline{u'_k u'_i} \frac{\partial \overline{u'_j}}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\
 & = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \overline{P'} \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \overline{u_j}}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \overline{u_i}}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\
 & = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \overline{P'} \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \overline{u_k u'_i u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\overline{u'_i u'_k} \frac{\partial \bar{u}_j}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\ &= - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} + \frac{\partial \overline{u'_j P'}}{\partial x_i} \right] + \bar{P}' \left[\frac{\partial \overline{u'_i}}{\partial x_j} + \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i \theta'} - g_i \overline{u'_j \theta'}) + \nu \frac{\partial}{\partial x_k} \left(\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right) - 2\nu \left(\frac{\partial \overline{u'_j}}{\partial x_k} \frac{\partial \overline{u'_i}}{\partial x_k} \right) \end{aligned}$$

$$\begin{aligned} & \frac{\partial \overline{u'_i u'_j}}{\partial t} + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{u'_i u'_j} + \overline{u'_k u'_i u'_j} - \nu \frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right] + \frac{\partial \overline{P' u'_i}}{\partial x_j} + \frac{\partial \overline{P' u'_j}}{\partial x_i} + f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) \\ &= - \overline{u'_k u'_i} \frac{\partial \bar{u}_j}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial \bar{u}_i}{\partial x_k} - \beta (g_j \overline{u'_i \theta'} + g_i \overline{u'_j \theta'}) + \overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} - 2\nu \overline{\left(\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k} \right)} \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\frac{\partial u'_j}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k u'_j + \bar{u}_j u'_k + u'_k u'_j - \overline{u'_k u'_j}) + \epsilon_{j,k,l} f_k u'_l = -\frac{\partial P'}{\partial x_j} - g_j \beta \theta' + \nu \nabla^2 u'_j \quad (5)$$

$$\frac{\partial \theta'}{\partial t} + \frac{\partial}{\partial x_k} (\bar{\theta} u'_k + \bar{u}_k \theta' + u'_k \theta' + \overline{u'_k \theta'}) = \alpha \nabla^2 \theta' \quad (6)$$

$$\frac{\partial u'_j \theta'}{\partial t} = u'_j \frac{\partial \theta'}{\partial t} + \theta' \frac{\partial u'_j}{\partial t}$$

$$\begin{aligned} \frac{\partial \overline{u'_j \theta'}}{\partial t} + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{\theta' u'_j} + \overline{u'_k u'_j \theta'} - \alpha \overline{u'_j \frac{\partial \theta'}{\partial x_k}} - \nu \overline{\theta' \frac{\partial u'_j}{\partial x_k}} \right] + \frac{\partial}{\partial x_j} \overline{P' \theta'} + \epsilon_{j,k,l} f_k \overline{u'_l \theta'} \\ = -\overline{u'_j u'_k} \frac{\partial \bar{\theta}}{\partial x_k} - \overline{\theta' u'_k} \frac{\partial \bar{u}_j}{\partial x_k} - \beta g_j \overline{\theta'^2} + \overline{P' \frac{\partial \theta'}{\partial x_j}} - (\alpha + \nu) \overline{\frac{\partial u'_j}{\partial x_k} \frac{\partial \theta'}{\partial x_k}} \end{aligned} \quad (8)$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

A equação 8 envolve $\overline{\theta'^2}$, na equação . Onde é necessário obter uma equação para este termo da equação 6

$$\frac{\partial \theta'}{\partial t} + \frac{\partial}{\partial x_k} (\bar{\theta} u'_k + \bar{u}_k \theta' + u'_k \theta' + \overline{u'_k \theta'}) = \alpha \nabla^2 \theta' \quad (6)$$

$$\frac{\partial \overline{\theta'^2}}{\partial t} = + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{\theta'^2} + \overline{k'_k \theta'^2} - \alpha \frac{\partial \overline{\theta'^2}}{\partial x_k} \right] = 2 \overline{k'_k \theta'^2} - 2 \alpha \frac{\partial \theta'}{\partial x_k} \frac{\partial \theta'}{\partial x_k} \quad (9)$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

Suposições de modelagem

. A principal contribuição de **Rotta (1951)** foi sugerir uma suposição para o termo, $\overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)}$, que ele chamou de "termo de redistribuição de energia", uma vez que uma de suas funções é particionar a energia entre os três componentes de energia sem contribuir para o total. contração, o termo sai da Ed.7. Com base nas relações integrais obtidas da Eq.5 no caso neutro onde $-\beta(g_j u'_i \theta')$ não é significativa, Rotta sugeriu que o termo poderia ser razoavelmente proporcional a $u'_i u'_j$ e $\frac{\partial \bar{u}_i}{\partial x_j}$. Por isso,

$$\overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} = C_{ijk m} \overline{u'_k u'_m} + C'_{ijk m} \frac{\partial \bar{u}_k}{\partial x_m}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

Suposições de modelagem

aqui assumimos que os coeficientes constitutivos são tensores isotrópicos, ou seja

$$C_{ijklm} = C_1 \delta_{i,j} \delta_{k,m} + C_2 \delta_{i,k} \delta_{j,m} + C_3 \delta_{j,k} \delta_{i,m}$$

Da eq. Da continuidade obtém-se $C_i = \frac{(C_2+C_3)}{3}$ raciocínio semelhante se aplica ao C_{ijklm} . Obtém-se portanto

$$\overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} = C_{ijklm} \overline{u'_k u'_m} + C'_{ijklm} \frac{\partial \bar{u}_k}{\partial x_m}$$

$$\overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} = -\frac{q}{3l_1} \left(\overline{u'_i u'_j} - \frac{\delta_{i,j}}{3} q^2 + C q^2 \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \right)$$

Onde $-\frac{q}{3l_1}$ e Cq^2 the sido substituído por coeficientes escalares sobreviventes $q \equiv \left(\overline{u_i'^2} \right)^{1/2}$. O comprimento l_1 e a constante C devem ser determinadas empiricamente.



Linearização da Equações de Navier Stokes



5.2 TURBULENT KINETIC ENERGY

A equação de Navier Stokes: conservação de momentum

$$\begin{aligned} & \frac{\partial(u_i' u_k')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i' u_k')}{\partial x_j} \\ &= -(u_j' u_k') \frac{\partial(\bar{u}_i)}{\partial x_j} - (u_j' \bar{u}_i) \frac{\partial(u_k')}{\partial x_j} - \bar{u}_i u_k' \frac{\partial(u_j')}{\partial x_j} - \frac{\partial(u_j' u_i' u_k')}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(P' u_k')}{\partial x_i} - g \frac{\rho' u_k'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k' u_k') + \nu \frac{\partial^2(u_i' u_k')}{\partial x_j^2} \end{aligned}$$

Considere que a propriedade da equação da continuidade para o fluxo $\nabla \cdot \vec{V} = 0$

$$\nabla \cdot \vec{V}' = \frac{\partial(u_j')}{\partial x_j} = 0, \quad j = 1, 2, 3$$

$$\frac{\partial(u_i' u_k')}{\partial t} + (\bar{u}_j) \frac{\partial(u_i' u_k')}{\partial x_j} = -(u_j' u_k') \frac{\partial(\bar{u}_i)}{\partial x_j} - (u_j' \bar{u}_i) \frac{\partial(u_k')}{\partial x_j} - \frac{\partial(u_j' u_i' u_k')}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(P' u_k')}{\partial x_i} - g \frac{\rho' u_k'}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (u_k' u_k') + \nu \frac{\partial^2(u_i' u_k')}{\partial x_j^2}$$



Linearização da Equações de Navier Stokes



$$\begin{aligned} & \frac{\partial \overline{u'_j \theta'}}{\partial t} + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{\theta' u'_j} + \overline{u'_k u'_j \theta'} - \alpha \overline{u'_j \frac{\partial \theta'}{\partial x_k}} - \nu \overline{\theta' \frac{\partial u'_j}{\partial x_k}} \right] + \frac{\partial}{\partial x_j} \overline{P' \theta'} + \epsilon_{j,k,l} f_k \overline{u'_l \theta'} \\ &= -\overline{u'_j u'_k} \frac{\partial \bar{\theta}}{\partial x_k} - \overline{\theta' u'_k} \frac{\partial \bar{u}_j}{\partial x_k} - \beta g_j \overline{\theta'^2} + \overline{P' \frac{\partial \theta'}{\partial x_j}} - (\alpha + \nu) \overline{\frac{\partial u'_j}{\partial x_k} \frac{\partial \theta'}{\partial x_k}} \end{aligned} \quad (8)$$

$$\begin{aligned} & \frac{\partial \overline{u'_i u'_j}}{\partial t} + \frac{\partial}{\partial x_k} \left[\bar{u}_k \overline{u'_i u'_j} + \overline{u'_k u'_i u'_j} - \nu \overline{\frac{\partial u'_i u'_j}{\partial x_k}} \right] + \frac{\partial \overline{P' u'_i}}{\partial x_j} + \frac{\partial \overline{P' u'_j}}{\partial x_i} + f_k (\epsilon_{j,k,l} u'_l \textcolor{red}{u'_i} + \epsilon_{\textcolor{teal}{i},k,l} u'_l \textcolor{teal}{u'_j}) \\ &= -\overline{u'_k u'_i} \frac{\partial \bar{u}_j}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial \bar{u}_i}{\partial x_k} - \beta (g_j \overline{u'_i \theta'} + g_i \overline{u'_j \theta'}) + \overline{P' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} - 2\nu \overline{\left(\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k} \right)} \end{aligned}$$



Linearização da Equações de Navier Stokes

5.2 TURBULENT KINETIC ENERGY

A equação de Navier Stokes: conservação de momentum



$$\begin{aligned}
& \frac{\partial \overline{u'_i u'_j}}{\partial t} + \frac{\partial}{\partial x_k} [\bar{u}_k \overline{u'_i u'_j}] \\
&= -\overline{u'_k u'_i} \frac{\partial \bar{u}_j}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial \bar{u}_i}{\partial x_k} - \frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} - \beta (g_j \overline{u'_i \theta'} + g_i \overline{u'_j \theta'}) - f_k (\epsilon_{j,k,l} \overline{u'_l u'_i} + \epsilon_{i,k,l} \overline{u'_l u'_j}) - \left[\frac{\partial \overline{P' u'_i}}{\partial x_j} + \frac{\partial \overline{P' u'_j}}{\partial x_i} \right] - 2\nu \left(\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k} \right) \\
&+ \nu \frac{\partial}{\partial x_k} \left[\frac{\partial \overline{u'_i u'_j}}{\partial x_k} \right] + \overline{P'} \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)
\end{aligned}$$

O termo do lado esquerdo é a taxa temporal local de mudança e advecção de $\overline{(u'_i u'_k)}$

O **1 e termo** do lado direito são os termos de produção resultante da interação da turbulência e o escoamento médio

O **2 termo (terceiro momento)** correlação tripla pode ser interpretado como transporte de turbulência (segundo momento) pela flutuação turbulenta com o ganho ou perda devido a divergência do fluxo turbulento

O **3 termo** representa a produção e destruição da flutuabilidade (conversão da energia cinética turbulenta para a energia potencial turbulenta)

O **termo 4** é a rotação e pode ser desprezado para média temporal menor do que 1 hora

O **termo 5** é a interação da flutuação de pressão e do campo de velocidade

O **termo 6** é a dissipação molecular

O termo 7 é a redistribuição de energia, uma vez que uma de suas funções é particionar a energia entre os três componentes de energia sem contribuir para o total



Linearização da Equações de Navier Stokes

Para caso de homogeneidade horizontal

$$\begin{aligned} & \frac{\partial \overline{(u_i' u_k')}}{\partial t} + (\bar{u}_j) \frac{\partial \overline{(u_i' u_k')}}{\partial x_j} \\ &= -\overline{(u_j' u_k')} \frac{\partial (\bar{u}_i)}{\partial x_j} - \frac{\partial \overline{(u_j' u_i' u_k')}}{\partial x_j} - g \frac{\overline{\rho' u_k'}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j \overline{(u_k' u_k')} - \frac{1}{\rho_0} \frac{\partial \overline{(P' u_k')}}{\partial x_i} + \nu \frac{\partial^2 \overline{(u_i' u_k')}}{\partial x_j^2} \end{aligned}$$

O termo molecular $\nu \frac{\partial^2 \overline{(u_i' u_k')}}{\partial x_j^2} \rightarrow 0$ é desprezado no caso da covariância, porque a viscosidade é dominante somente em numero de ondas grandes.

Porém, neste caso a turbulência é isotrópica e assim a covariância é zero na horizontal.

$$(\mathbf{j}=\mathbf{k}) \quad \frac{\partial \overline{u' w'}}{\partial t} = -\overline{w'^2} \frac{\partial \bar{u}}{\partial z} + \frac{g}{\theta_v} \overline{u' \theta_v'} - \frac{\partial \overline{u' w'^2}}{\partial z} - \frac{1}{\rho} \left(\overline{u' \frac{\partial P'}{\partial z}} + \overline{w' \frac{\partial P'}{\partial x}} \right)$$

Isotrópico é a caracterização de uma substância que possui as mesmas propriedades físicas, independentemente da direção considerada.



Linearização da Equações de Navier Stokes

Para caso de homogeneidade horizontal e o estado básico em condições neutra $\frac{\partial \overline{u'w'}}{\partial t} = 0$ e $\frac{g}{\theta_v} \overline{u'\theta'_v} = 0$

$$(\mathbf{j}=\mathbf{k}) \quad \frac{\partial \overline{u'w'}}{\partial t} = -\overline{w'^2} \frac{\partial \bar{u}}{\partial z} + \frac{g}{\theta_v} \overline{u'\theta'_v} - \frac{\partial \overline{u'w'^2}}{\partial z} - \frac{1}{\rho} \left(\overline{u' \frac{\partial P'}{\partial z}} + \overline{w' \frac{\partial P'}{\partial x}} \right)$$

Isto mostra que o termo de correlação de pressão-velocidade destrói o stress na mesma taxa como ela é produzida

$$0 = -\overline{w'^2} \frac{\partial \bar{u}}{\partial z} - \frac{\partial \overline{u'w'^2}}{\partial z} - \frac{1}{\rho} \left(\overline{u' \frac{\partial P'}{\partial z}} + \overline{w' \frac{\partial P'}{\partial x}} \right)$$

$$+\overline{w'^2} \frac{\partial \bar{u}}{\partial z} + \frac{\partial \overline{u'w'^2}}{\partial z} = -\frac{1}{\rho} \left(\overline{u' \frac{\partial P'}{\partial z}} + \overline{w' \frac{\partial P'}{\partial x}} \right)$$



Equação da Energia Cinética Turbulenta



Linearização da Equações de Navier Stokes



5.2 Energia Cinética Turbulenta

$$\begin{aligned} & \frac{\partial \overline{(u_i' u_k')}}{\partial t} + (\bar{u}_j) \frac{\partial \overline{(u_i' u_k')}}{\partial x_j} \\ &= -\overline{(u_j' u_k')} \frac{\partial (\bar{u}_i)}{\partial x_j} - \frac{\partial \overline{(u_j' u_i' u_k')}}{\partial x_j} - g \frac{\overline{\rho' u_k'}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j \overline{(u_k' u_k')} - \frac{1}{\rho_0} \frac{\partial \overline{(P' u_k')}}{\partial x_i} + \nu \frac{\partial^2 \overline{(u_i' u_k')}}{\partial x_j^2} \end{aligned}$$

$$\overline{u_k' u_k'} = \overline{u_k'^2} = \overline{u_k'}^2 = 0$$

$$\bar{e} = \frac{\overline{u_i'^2}}{2} = \frac{\overline{(u'^2 + v'^2 + w'^2)}}{2} \quad i = k = 1, 2, 3$$

$$\frac{\partial \bar{e}}{\partial t} + (\bar{u}_j) \frac{\partial \bar{e}}{\partial x_j} = -\overline{(u_j' u_i')} \frac{\partial (\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{\rho_0} \delta_{i3} - \frac{\partial \overline{(e u_j')}}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial \overline{(P' u_i')}}{\partial x_i} + \nu \frac{\partial^2 \overline{(u_i' u_k')}}{\partial x_j^2}$$

$$\frac{\partial \bar{e}}{\partial t} + (\bar{u}_j) \frac{\partial \bar{e}}{\partial x_j} = -\frac{\overline{(u_j' u_i')}}{2} \frac{\partial (\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{2\rho_0} \delta_{i3} - \frac{\partial \overline{(e u_j')}}{\partial x_j} - \frac{1}{2\rho_0} \frac{\partial \overline{(P' u_i')}}{\partial x_i} - \epsilon$$



Linearização da Equações de Navier Stokes

5.2 Energia Cinética Turbulenta

$$\frac{\partial \bar{e}}{\partial t} + (\bar{u}_j) \frac{\partial \bar{e}}{\partial x_j} = - \frac{\overline{(u_j' u_i')}}{2} \frac{\partial (\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{2 \rho_0} \delta_{i3} - \frac{\partial (\overline{e u_j'})}{\partial x_j} - \frac{1}{2 \rho_0} \frac{\partial (\overline{P' u_i'})}{\partial x_i} - \epsilon$$

A quantidade ϵ é um parâmetro significativo para a atmosfera desde que seja relacionado a dissipação da energia cinética turbulenta de todos os movimentos atmosféricos



5.2 Energia Cinética Turbulenta

A essência da equação da energia cinética turbulenta pode ser expressa pela equação:

$$\frac{D\bar{e}}{Dt} = -\frac{\overline{(u_j' u_i')}}{2} \frac{\partial(\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{2\rho_0} \delta_{i3} - \frac{\partial(\overline{e u_j'})}{\partial x_j} - \frac{1}{2\rho_0} \frac{\partial(\overline{P' u_i'})}{\partial x_i} - \epsilon$$

$$\frac{\bar{D}(TKE)}{Dt} = MP + BPL + TR - \epsilon$$

MP é a produção mecânica

BPL é a produção e perda por flutuabilidade

TR redistribuição de tke por transporte e força de pressão

ϵ dissipação por atrito

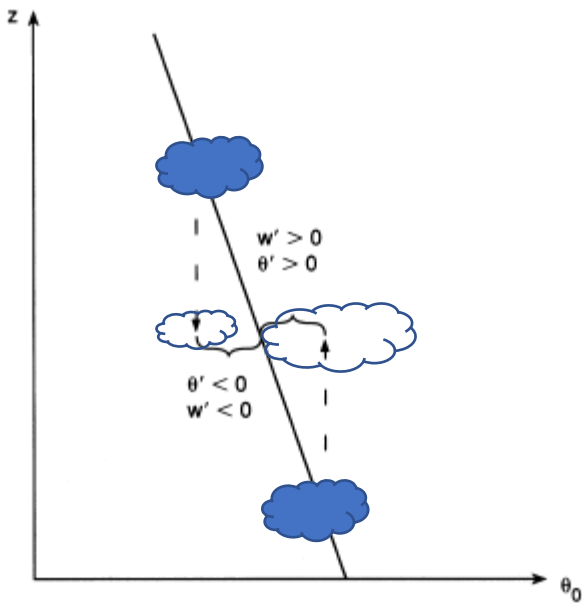


5.2 Energia Cinética Turbulenta

$$BPL \equiv \overline{w' \theta'} \frac{g}{\theta_0}$$

É a conversão da energia potencial do escoamento médio e a energia cinética turbulenta:
É positivo para movimentos que baixa o centro de massa da atmosfera
É negativo para movimentos que aumenta o centro de massa da atmosfera

Correlação positiva(fonte tke)
Atms. instável



Correlação negativa (destroi tke)
Atms. estável

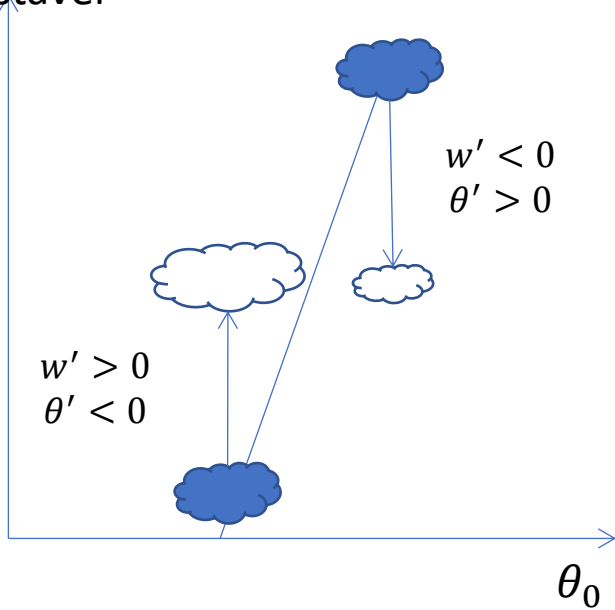


Fig. 5.1 Correlation between vertical velocity and potential temperature perturbations for upward or downward parcel displacements when the mean potential temperature $\theta_0(z)$ decreases with height.



5.2 Energia Cinética Turbulenta

Para ambos as condições estáveis e instáveis da CLP a turbulência pode ser produzida mecanicamente pela instabilidade dinâmica através do cisalhamento. Conversão de energia entre o escoamento médio e a flutuação turbulenta.

$$MP \equiv -\frac{\overline{u'w'}}{2} \frac{\partial \bar{u}}{\partial z} - \frac{\overline{v'w'}}{2} \frac{\partial \bar{v}}{\partial z}$$

$MP > 0$ quando o fluxo de momentum ($\overline{u'w'}$) é direcionado para baixo e o gradiente vertical é positivo



5.2 Energia Cinética Turbulenta

Estatisticamente na camada limite estável a turbulência pode existir somente se a produção mecânica for grande o suficiente para superar o efeito de supressão da estabilidade e da viscosidade

Esta condição é medida pelo numero de Richardson de fluxo

$$Rf \equiv -\frac{BPL}{MP} \equiv \frac{\overline{w'\theta' \frac{g}{\theta_0}}}{-\overline{u'w'} \frac{\partial \bar{u}}{\partial z} - \overline{v'w'} \frac{\partial \bar{v}}{\partial z}}$$



5.2 Energia Cinética Turbulenta

$$Rf \equiv -\frac{BPL}{MP} \equiv \frac{\overline{w'\theta' \frac{g}{\theta_0}}}{-\overline{u'w'} \frac{\partial \bar{u}}{\partial z} - \overline{v'w'} \frac{\partial \bar{v}}{\partial z}}$$

- **Se $Rf < 0$ a CLP é estatisticamente instável (a turbulências é sustentada pela convecção)**
- **$Rf > 0$ a CLP é estatisticamente estável**
- **$Rf < 0.25$ (a produção mecânica excede a produção por flutuabilidade por um fato de 4)**



5.2 Energia Cinética Turbulenta

5.3 Equações de momentum da camada limite Planetária

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} = - \frac{\partial(\overline{u_j' u_i'})}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\bar{\rho}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

$$\begin{aligned} & \frac{\partial(\overline{u_i' u_k'})}{\partial t} + (\bar{u}_j) \frac{\partial(\overline{u_i' u_k'})}{\partial x_j} \\ &= - \overline{(u_j' u_k')} \frac{\partial(\bar{u}_i)}{\partial x_j} - g \frac{\overline{\rho' u_k'}}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j \overline{(u_k' u_k')} - \frac{\partial(\overline{u_j' u_i' u_k'})}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\overline{P' u_k'})}{\partial x_i} + \nu \frac{\partial^2(\overline{u_i' u_k'})}{\partial x_j^2} \end{aligned}$$

$$\frac{\partial \bar{e}}{\partial t} + (\bar{u}_j) \frac{\partial \bar{e}}{\partial x_j} = - \frac{\overline{(u_j' u_i')}}{2} \frac{\partial(\bar{u}_i)}{\partial x_j} - g \frac{\overline{u_i' \rho'}}{2\rho_0} \delta_{i3} - \frac{\partial(\overline{e u_j'})}{\partial x_j} - \frac{1}{2\rho_0} \frac{\partial(\overline{P' u_i'})}{\partial x_i} - \epsilon$$



5.2 Energia Cinética Turbulenta

5.3 Equações de momentum da camada limite Planetária

$$\frac{\partial(\bar{u}_i)}{\partial t} + (\bar{u}_j) \frac{\partial(\bar{u}_i)}{\partial x_j} = - \frac{\partial(u_j' u_i')}{\partial x_j} - \frac{1}{\rho_0} \frac{\partial(\bar{P})}{\partial x_i} - g \frac{\rho}{\rho_0} \delta_{i3} - 2\Omega \varepsilon_{ijk} \eta_j (\bar{u}_k) + \nu \frac{\partial^2(\bar{u}_i)}{\partial x_j^2}$$

Para o caso especial de turbulência horizontalmente homogênea:

-> A camada viscosa, a viscosidade molecular e o termo da divergência horizontal do fluxo de momentum turbulento podem ser desprezados.

$$\frac{D\bar{u}}{Dt} = - \frac{1}{\rho_0} \frac{\partial \bar{P}}{\partial x} + f\bar{v} - \frac{\partial \overline{u'w'}}{\partial z}$$

$$\frac{D\bar{v}}{Dt} = - \frac{1}{\rho_0} \frac{\partial \bar{P}}{\partial y} - f\bar{u} - \frac{\partial \overline{v'w'}}{\partial z}$$

Só pode ser resolvida se conhecermos a distribuição vertical do fluxo de momentum



fim



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\overline{u'_i} \frac{\partial u'_j}{\partial t} + \overline{u'_j} \frac{\partial u'_i}{\partial t} \right) + \left(\overline{u'_i} \frac{\partial \bar{u}_k u'_j}{\partial x_k} + \overline{u'_j} \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right) + \left(\overline{u'_i} \frac{\partial u'_k u'_j}{\partial x_k} + \overline{u'_j} \frac{\partial u'_k u'_i}{\partial x_k} \right) + \left(\overline{u'_i} \frac{\partial \bar{u}_j u'_k}{\partial x_k} + \overline{u'_j} \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right) - \left(\overline{u'_i} \frac{\partial \bar{u}'_k u'_j}{\partial x_k} + \overline{u'_j} \frac{\partial \bar{u}'_k u'_i}{\partial x_k} \right) \\ & + \epsilon_{j,k,l} f_k u'_l \overline{u'_i} + \epsilon_{i,k,l} f_k u'_l \overline{u'_j} = -\overline{u'_i} \frac{\partial P'}{\partial x_j} - \overline{u'_j} \frac{\partial P'}{\partial x_i} - g_j \beta \overline{u'_i} \theta' - g_i \beta \overline{u'_j} \theta' + \nu \overline{u'_i} \nabla^2 u'_j + \nu \overline{u'_j} \nabla^2 u'_i \end{aligned}$$

Aplica-se as derivadas anteriores

$$\begin{aligned} \Rightarrow \frac{\partial (\bar{u}_k u'_i) u'_j}{\partial x_k} &= u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} + \bar{u}_k u'_i \frac{\partial u'_j}{\partial x_k} & \Rightarrow \frac{\partial (u'_k u'_i) u'_j}{\partial x_k} &= u'_j \frac{\partial u'_k u'_i}{\partial x_k} + u'_k u'_i \frac{\partial u'_j}{\partial x_k} \\ \Rightarrow \frac{\partial (\bar{u}_k u'_j) u'_i}{\partial x_k} &= \overline{u'_i} \frac{\partial \bar{u}_k u'_j}{\partial x_k} + \bar{u}_k u'_j \frac{\partial \overline{u'_i}}{\partial x_k} & \Rightarrow \frac{\partial (u'_k u'_j) u'_i}{\partial x_k} &= u'_i \frac{\partial u'_k u'_j}{\partial x_k} + u'_k u'_j \frac{\partial u'_i}{\partial x_k} \end{aligned}$$

$$\begin{aligned} & \left(\frac{\partial \overline{u'_i u'_j}}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \overline{u'_i u'_j}}{\partial x_k} - \overline{u'_j} \frac{\partial \bar{u}_k \overline{u'_i}}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_k \overline{u'_j u'_i}}{\partial x_k} - \overline{u'_i} \frac{\partial \bar{u}_k \overline{u'_j}}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \overline{u'_k u'_i u'_j}}{\partial x_k} - \overline{u'_j} \frac{\partial \overline{u'_k u'_i}}{\partial x_k} \right] + \left[\frac{\partial \overline{u'_k u'_j u'_i}}{\partial x_k} - \overline{u'_i} \frac{\partial \overline{u'_k u'_j}}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j \overline{u'_i u'_k}}{\partial x_k} - \overline{u'_k} \frac{\partial \bar{u}_j \overline{u'_i}}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_i \overline{u'_j u'_k}}{\partial x_k} - \overline{u'_k} \frac{\partial \bar{u}_i \overline{u'_j}}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \overline{u'_i u'_k u'_j}}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial \overline{u'_i}}{\partial x_k} \right] + \left[\frac{\partial \overline{u'_j u'_k u'_i}}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial \overline{u'_j}}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l \overline{u'_i} + \epsilon_{i,k,l} u'_l \overline{u'_j}) = - \left[\frac{\partial \overline{u'_i P'}}{\partial x_j} - \overline{P'} \frac{\partial \overline{u'_i}}{\partial x_j} \right] - \left[\frac{\partial \overline{u'_j P'}}{\partial x_i} - \overline{P'} \frac{\partial \overline{u'_j}}{\partial x_i} \right] - \beta (g_j \overline{u'_i} \theta' - g_i \overline{u'_j} \theta') + \nu \overline{u'_i} \nabla^2 u'_j + \nu \overline{u'_j} \nabla^2 u'_i \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} - u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} - u'_j \frac{\partial u'_k u'_i}{\partial x_k} \right] + \left[\frac{\partial u'_k u'_j u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} - u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} \right] + \left[\frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} - \overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} - \overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} - P' \frac{\partial u'_i}{\partial x_j} \right] - \left[\frac{\partial u'_j P'}{\partial x_i} - P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$

Agrupe os termos

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} + \frac{\partial u'_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial u'_k u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} + \frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] - \left(\left[-\overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[-\overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + \left[P' \frac{\partial u'_i}{\partial x_j} + P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \frac{\partial \bar{u}_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} + \frac{\partial u'_k u'_j u'_i}{\partial x_k} \right] + \left[-u'_j \frac{\partial u'_k u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[-\overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + \left[P' \frac{\partial u'_i}{\partial x_j} + P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$

Aplica-se as derivadas anteriores

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] + \left[-u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} + \frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] + \left[-u'_j \frac{\partial u'_k u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\overline{u'_k u'_j} \frac{\partial u'_i}{\partial x_k} \right] + \left[-\overline{u'_k u'_i} \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + \left[P' \frac{\partial u'_i}{\partial x_j} + P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] + \left[-u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k} - u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} + \frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] + \left[-u'_j \frac{\partial u'_k u'_i}{\partial x_k} - u'_i \frac{\partial u'_k u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_j u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} \right] + \left[-\bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + \left[P' \frac{\partial u'_i}{\partial x_j} + P' \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \nabla^2 u'_j + \nu u'_j \nabla^2 u'_i \end{aligned}$$

$$-\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} + \bar{u}_k \frac{\partial u'_i u'_j}{\partial x_k} = -u'_i \frac{\partial \bar{u}_k u'_j}{\partial x_k} - u'_j \frac{\partial \bar{u}_k u'_i}{\partial x_k}$$

$$-\frac{\partial u'_k u'_i u'_j}{\partial x_k} + u'_k \frac{\partial u'_i u'_j}{\partial x_k} = -u'_i \frac{\partial u'_k u'_j}{\partial x_k} - u'_j \frac{\partial u'_k u'_i}{\partial x_k}$$

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] - \left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} - \bar{u}_k \frac{\partial u'_i u'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] - \left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} - u'_k \frac{\partial u'_i u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_j u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \frac{\partial}{\partial x_k} \frac{\partial u'_j}{\partial x_k} + \nu u'_j \frac{\partial}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(2 \left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] - \left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} - \bar{u}_k \frac{\partial u'_i u'_j}{\partial x_k} \right] \right) + \left(2 \left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] - \left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} - u'_k \frac{\partial u'_i u'_j}{\partial x_k} \right] \right) \\ & + \left(\left[\frac{\partial \bar{u}_j u'_i u'_k}{\partial x_k} + \frac{\partial \bar{u}_i u'_j u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) \\ & + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu u'_i \frac{\partial}{\partial x_k} \frac{\partial u'_j}{\partial x_k} + \nu u'_j \frac{\partial}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \end{aligned}$$

$$\begin{aligned} & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \bar{u}_k \frac{\partial u'_i u'_j}{\partial x_k} + u'_k \frac{\partial u'_i u'_j}{\partial x_k} \\ & + \left(\left[u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} + u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + \bar{u}_j \frac{\partial u'_i u'_k}{\partial x_k} + u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} + \bar{u}_i \frac{\partial u'_j u'_k}{\partial x_k} \right] + \left[-u'_k \frac{\partial \bar{u}_j u'_i}{\partial x_k} - u'_k \frac{\partial \bar{u}_i u'_j}{\partial x_k} \right] \right) \\ & - \left(\left[\frac{\partial u'_i u'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j u'_k u'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\ & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \left[\frac{\partial}{\partial x_k} \left(u'_i \frac{\partial u'_j}{\partial x_k} \right) - \left(\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k} \right) \right] \\ & + \nu \left[\frac{\partial}{\partial x_k} \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) - \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right) \right] \end{aligned}$$





1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[+\mathbf{u}'_i \frac{\partial \bar{u}_j \mathbf{u}'_k}{\partial x_k} + \mathbf{u}'_j \frac{\partial \bar{u}_i \mathbf{u}'_k}{\partial x_k} \right] + \left[\mathbf{u}'_k \frac{\partial \bar{u}_j \mathbf{u}'_i}{\partial x_k} - \mathbf{u}'_k \frac{\partial \bar{u}_j \mathbf{u}'_i}{\partial x_k} + \mathbf{u}'_k \frac{\partial \bar{u}_i \mathbf{u}'_j}{\partial x_k} - \mathbf{u}'_k \frac{\partial \bar{u}_i \mathbf{u}'_j}{\partial x_k} \right] \right) \\
 & - \left(\left[\frac{\partial \mathbf{u}'_i \mathbf{u}'_k \mathbf{u}'_j}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \mathbf{u}'_k \mathbf{u}'_i}{\partial x_k} \right] \right) - \left(\left[-\bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) \\
 & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \left[\frac{\partial}{\partial x_k} \left(\mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) - \left(\frac{\partial \mathbf{u}'_i}{\partial x_k} \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) \right] \\
 & + \nu \left[\frac{\partial}{\partial x_k} \left(\mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) - \left(\frac{\partial \mathbf{u}'_j}{\partial x_k} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) \right]
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\mathbf{u}'_i \frac{\partial \bar{u}_j \mathbf{u}'_k}{\partial x_k} + \mathbf{u}'_j \frac{\partial \bar{u}_i \mathbf{u}'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \mathbf{u}'_k \mathbf{u}'_j}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \mathbf{u}'_k \mathbf{u}'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) \\
 & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(\mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) + \left(\mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial \mathbf{u}'_j}{\partial x_k} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i \frac{\partial \bar{u}_j u'_k}{\partial x_k} + u'_j \frac{\partial \bar{u}_i u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial u'_i u'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k u'_i u'_j}{\partial x_k} \right] \right) + \left(\left[u'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + u'_i \bar{u}_j \frac{\partial u'_k}{\partial x_k} + u'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} + u'_j \bar{u}_i \frac{\partial u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial u'_i \bar{u}'_k u'_j}{\partial x_k} \right] + \left[\frac{\partial u'_j \bar{u}'_k u'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k u'_j \frac{\partial u'_i}{\partial x_k} - \bar{u}'_k u'_i \frac{\partial u'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l u'_i + \epsilon_{i,k,l} u'_l u'_j) \\
 & = - \left[\frac{\partial u'_i P'}{\partial x_j} + \frac{\partial u'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right] - \beta (g_j u'_i \theta' - g_i u'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(u'_i \frac{\partial u'_j}{\partial x_k} \right) + \left(u'_j \frac{\partial u'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial u'_j}{\partial x_k} \frac{\partial u'_i}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} + \mathbf{u}'_i \bar{u}_j \frac{\partial \mathbf{u}'_k}{\partial x_k} + \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} + \mathbf{u}'_j \bar{u}_i \frac{\partial \mathbf{u}'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) \\
 & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(\mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) + \left(\mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial \mathbf{u}'_j}{\partial x_k} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial \mathbf{u}'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\mathbf{u}'_i \mathbf{u}'_k \frac{\partial \bar{u}_j}{\partial x_k} + \mathbf{u}'_j \mathbf{u}'_k \frac{\partial \bar{u}_i}{\partial x_k} + \mathbf{u}'_i \bar{u}_j \frac{\partial \mathbf{u}'_k}{\partial x_k} + \mathbf{u}'_j \bar{u}_i \frac{\partial \mathbf{u}'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} \mathbf{u}'_l \mathbf{u}'_i + \epsilon_{i,k,l} \mathbf{u}'_l \mathbf{u}'_j) \\
 & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(\mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) + \left(\mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial \mathbf{u}'_j}{\partial x_k} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right)
 \end{aligned}$$



1.1 A Teoria de Similaridade de Monin-Obukhov



2. As equações básicas

$$\begin{aligned}
 & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\mathbf{u}'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + \mathbf{u}'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} + \mathbf{u}'_i \bar{u}_j \frac{\partial u'_k}{\partial x_k} + \mathbf{u}'_j \bar{u}_i \frac{\partial u'_k}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) \\
 & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(\mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) + \left(\mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial \mathbf{u}'_j}{\partial x_k} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right)
 \end{aligned}$$

$$\begin{aligned}
 & \left(\frac{\partial \mathbf{u}'_i \mathbf{u}'_j}{\partial t} \right) + \left(\left[\frac{\partial \bar{u}_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\frac{\partial u'_k \mathbf{u}'_i \mathbf{u}'_j}{\partial x_k} \right] \right) + \left(\left[\mathbf{u}'_i u'_k \frac{\partial \bar{u}_j}{\partial x_k} + \mathbf{u}'_j u'_k \frac{\partial \bar{u}_i}{\partial x_k} \right] \right) - \left(\left[\frac{\partial \mathbf{u}'_i \bar{u}'_k \mathbf{u}'_j}{\partial x_k} \right] + \left[\frac{\partial \mathbf{u}'_j \bar{u}'_k \mathbf{u}'_i}{\partial x_k} \right] \right) \\
 & - \left(\left[-\bar{u}'_k \mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} - \bar{u}'_k \mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right] \right) + f_k (\epsilon_{j,k,l} u'_l \mathbf{u}'_i + \epsilon_{i,k,l} u'_l \mathbf{u}'_j) \\
 & = - \left[\frac{\partial \mathbf{u}'_i P'}{\partial x_j} + \frac{\partial \mathbf{u}'_j P'}{\partial x_i} \right] + P' \left[\frac{\partial \mathbf{u}'_i}{\partial x_j} + \frac{\partial \mathbf{u}'_j}{\partial x_i} \right] - \beta (g_j \mathbf{u}'_i \theta' - g_i \mathbf{u}'_j \theta') + \nu \frac{\partial}{\partial x_k} \left[\left(\mathbf{u}'_i \frac{\partial \mathbf{u}'_j}{\partial x_k} \right) + \left(\mathbf{u}'_j \frac{\partial \mathbf{u}'_i}{\partial x_k} \right) \right] - 2\nu \left(\frac{\partial \mathbf{u}'_j}{\partial x_k} \frac{\partial \mathbf{u}'_i}{\partial x_k} \right)
 \end{aligned}$$